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PrimeTime Workshop

Lab Guide

10-I-034-SLG-014 2016.06

Synopsys Customer Education Services
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Mountain View, California 94043

Workshop Registration: http://training.synopsys.com

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PrimeTime Lab Guide

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1 Generating Setup and Hold Reports

Learning Objectives

After completing this lab, you should be able to:

- Restore a previously saved PrimeTime session
- Take advantage of helpful PrimeTime commands that will make you more efficient when using PrimeTime interactively and show you how to find more information on commands, variables and your design library
- Interpret key components of a timing report for setup and hold timing checks

Lab Duration: 30 minutes

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Overview

Relevant Files and Directories

All files for this lab are located in the `lab1-6_reports` directory under your home directory.

`lab1-6_reports/`	Current working directory
`orca_savesession/`	Session to restore for labs
`.synopsys_pt.setup`	PT setup file

Answers / Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question or would like to confirm your results, check the back portion of this lab.

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Instructions

Your goal is to start mastering timing reports as well as learn useful PrimeTime commands.

Task 1. Restore a PrimeTime Sesssion

Learn to invoke a previously saved PrimeTime session to perform STA. The first 6 labs of this workshop will be performed in the same directory.

1. Invoke PrimeTime from the **lab1-6_reports** workshop lab Unix directory.

```
unix% cd lab1-6_reports
unix% pt_shell
```

2. Restore a previously saved PrimeTime session. This step will read in the design netlist, libraries, and constraints. The design is now ready for analysis.

   Note: The argument **orca_savesession** below is a Unix directory. You will learn more about saving a PrimeTime session later in the workshop.

   Note: PrimeTime supports command, option, variable and file completion. Type a few letters and then hit the tab key.

```
pt_shell> restore_session orca_savesession
```

3. Execute **report_analysis_coverage**.

```
pt_shell> report_analysis_coverage
```

   Question 1. From the header at the top of this report, what is the name of the design under analysis?

   .....

   Question 2. How many timing check violations does **ORCA** have?

   .....

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Task 2. Explore Helpful PrimeTime Commands

1. Execute the following three history short cut commands:

```
pt_shell> history
pt_shell> !!
pt_shell> !2
```

   Question 3. Describe the difference between the last two history commands above.

   .....

2. Even more powerful are the command line editing capabilities.

   Use up and down arrows to scroll through the history event list as an alternative to the previous step.

   Type the following to see all the available key bindings (in the default emacs editing mode).

```
pt_shell> list_key_bindings
```

3. Explore the **page mode** alias; execute the following command, which will generate a report that scrolls off the screen:

```
pt_shell> report_timing -group [get_path_group *]
```

4. Turn on page mode.

```
pt_shell> page_on
pt_shell> !rep
```

5. Use the **space bar** to page through a long report. Quit from a long report in page mode by typing "**q**". If you want to turn off page mode, use the command alias **page_off**.

6. Explore another way to view screens of command results: the **view** Tcl procedure.

7. Send a timing report to a separate window with **view**.

```
pt_shell> view report_timing -group [get_path_group *]
```


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8. Find the command to restore a PrimeTime session and then display help information on this command.

```
pt_shell> help restore*
pt_shell> man restore_session
pt_shell> restore_session -help
```

Note: The following is an alternative way to display syntax help.

```
pt_shell> help -v restore_session
```

Question 4. From the last command above, does the command **`restore_session`** accept switches?

.....

9. The time unit in PrimeTime is determined by the main technology library.

To find the time unit for **ORCA**, first list all libraries in memory.

Note: The * in the following report indicates the main library.

```
pt_shell> list_lib
```

10. Generate a report for the main library which will state the time unit.

Note: Use copy and paste to avoid mistyping the lib name. The time unit is at the very top of the report.

```
pt_shell> report_lib cb13fs120_tsmc_max
```

Question 5. What is the time unit used for timing reports (as well as all other reports) for the **ORCA** design?

.....

Note: Do not forget to use "q" to quit from a long report in page mode and return to the `pt_shell` prompt without reading the entire report!

11. Display a more focused library report, displaying only units of the current design.

```
pt_shell> report_units
```


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Task 3. Analyze STA Reports

Generate and intepret two STA reports for setup and hold for **`SYS_CLK`**.

1. By default, the command **`report_timing`** generates a single report for each clock in a design.

Execute the following to display the clocks in **ORCA**:

```
pt_shell> report_clock
```

Question 6. How many clocks are in **ORCA**?

.....

2. Create a single, "short" timing report for setup for the clock **`SYS_CLK`**. Use command-line expansion (the tab key) to expand both the command AND the options `-group` and `-path`.

```
pt_shell> report_timing -group SYS_CLK -path short
```

Note: The lines containing the data path cells and their delays are removed from the data arrival section making this report "short".

Note: The above command generates a report for setup by default.

Question 7. There are at least 4 clues that this report is for setup and not for hold. How many can you identify?

.....

.....

Question 8. Identify the instance names of the start and end point flip-flops.

.....

Question 9. The clock skew for this timing path is 0.511ns; which two lines in the report can you use to calculate this?

.....

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Question 10. How does this clock skew affect slack (i.e. does the clock skew help or hurt slack)?

.....

Question 11. How large is the violation in comparison to the clock period?

.....

3. Generate a timing report for hold time.

The following is a short cut that will execute the last command in history starting with the letters "**rep**" and add the switch **-delay min** (which will generate a report for hold time).

```
pt_shell> !rep -delay min
```

Question 12. There are at least 4 clues that indicate this is a hold report and not a setup report. How many can you find?

.....

Question 13. How does the clock skew in this hold report affect slack (i.e. does the clock skew help or hurt slack)?

.....

4. Quit PrimeTime.

```
pt_shell> quit
```

This completes Lab 1. Return to lecture.

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Answers / Solutions

Question 1. From the header at the top of this report, what is the name of the design under analysis?

The design is **ORCA**.

Question 2. How many timing check violations does **ORCA** have?

There are 138 violations: 13 for setup, 71 for hold, 6 for removal, 9 for out_setup, and 39 for out_hold.

Question 3. Describe the difference between the last two history commands above.

The command **!2** repeats the 2nd command executed in history. The command **!!** repeats the last executed command in history.

Question 4. From the last command above, does the command **restore_session** accept switches?

Yes, there is one switch for this command. It accepts a session name, allowing a user to specify which of many saved sessions to restore.

Question 5. What is the time unit used for timing reports (as well as all other reports) for the **ORCA** design?

1ns.

Question 6. How many clocks are in **ORCA**?

There are 6 clocks in **ORCA**.

Question 7. There are at least 4 clues that this report is for setup and not for hold. How many can you identify?

The most glaring clue is the "library setup time" in the generated report. The more subtle clues are:

- The clock edges used for the data arrival and data required are 0ns and 8ns respectively (and not 0ns and 0ns as for hold time).
- The "**Path Type**" in the header is **max** which indicates that this report is for setup (a "**Path Type**" of **min** indicates a hold report).
- Finally, the data arrival time is <u>after</u> the data required time and the slack is violated (whereas if this was a report for hold, the slack would be met!)

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Question 8. Identify the instance names of the start and end point flip-flops.

```
Startpoint: I_ORCA_TOP/I_BLENDER/s3_op2_reg[18]
            (rising edge-triggered flip-flop clocked by SYS_CLK)
Endpoint: I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]
            (rising edge-triggered flip-flop clocked by SYS_CLK)
Path Group: SYS_CLK
Path Type: max
Min Clock Paths Derating Factor : 0.900
```

Question 9. The clock skew for this timing path is 0.511 ns. Which lines in the report can you use to calculate this?

```
Point                                                  Incr       Path
-----------------------------------------------------------------------
clock SYS_CLK (rise edge)                              0.000      0.000
clock network delay (propagated)                       3.247      3.247
I_ORCA_TOP/I_BLENDER/s3_op2_reg[18]/CP (sdnrb1)        0.000      3.247 r
I_ORCA_TOP/I_BLENDER/s3_op2_reg[18]/Q (sdnrb1)         0.516 &    3.763 r
...
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/D (sdnrb1)         8.242 &   12.004 f
data arrival time                                                12.004

clock SYS_CLK (rise edge)                              8.000      8.000
clock network delay (propagated)                       2.736     10.736
.....
```

Question 10. How does this clock skew affect slack (i.e. does the clock skew help or hurt slack)?

The clock skew hurts the slack for setup in this specific timing report. The clock latency to the start point flip-flop causes the data to arrive 0.511ns later causing a larger setup timing violation..

Question 11. How large is the violation in comparison to the clock period?

The clock period is 8ns. The violation is 0.987. It is approximately 12% of the clock period.

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Question 12. There are at least 4 clues that indicate this is a hold report and not a setup report. How many can you find?

The most glaring clue is the highlighted "library hold time" in the report below. The more subtle clues are:

- The clock edges used for the data arrival and data required are 0ns and 0ns respectively.
- The "**Path Type**" in the header is **min** which indicates that this report is for hold time.
- Finally, the data arrival time is <u>before</u> the data required time and the slack is violated.

Question 13. How does the clock skew for this hold report affect slack (i.e. does the clock skew help or hurt slack)?

From the report, you can see (after applying the clock network delay) that the capture clock edge arrives later than the launch clock edge. This hurts the hold slack, causing a violation.

```
Point                                                  Incr       Path
-----------------------------------------------------------------------
clock SYS_CLK (rise edge)                              0.000      0.000
clock network delay (propagated)                       2.360      2.360
I_ORCA_TOP/I_PARSER/out_bus_reg[10]/CP (sdcrq1)        0.000      2.360 r
I_ORCA_TOP/I_PARSER/out_bus_reg[10]/Q (sdcrq1)         0.353 &    2.713 r
...
I_ORCA_TOP/I_BLENDER/rem_green_reg/D (sdcrn1)          0.142 &    2.855 f
data arrival time                                                 2.855

clock SYS_CLK (rise edge)                              0.000      0.000
clock network delay (propagated)                       3.206      3.206
clock reconvergence pessimism                         -0.216      2.990
I_ORCA_TOP/I_BLENDER/rem_green_reg/CP (sdcrn1)                    2.990 r
library hold time                                      0.006      2.997
data required time                                                2.997
-----------------------------------------------------------------------

data required time                                                2.997
data arrival time                                                -2.855
-----------------------------------------------------------------------
slack (VIOLATED)                                                 -0.142
```


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2

Timing Arcs in a Timing Report

Learning Objectives

After completing this lab, you should be able to:

- Interpret timing arcs and transitions in a timing report
- Generate timing reports for a specific data transition at the end point
- Generate timing reports with input pins included
- Generate a report on library timing arcs

Lab Duration: 20 minutes

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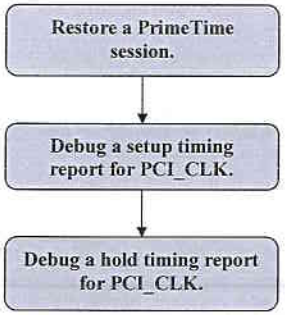
Overview



Relevant Files and Directories

All files for this lab are located in the `lab1-6_reports` directory under your home directory.

`lab1-6_reports/`	Current working directory
`orca_savesession/`	Session to restore for labs
`.synopsys_pt.setup`	PT setup file

Answers & Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question or would like to confirm your results, check the back portion of this lab.

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Instructions

Task 1. Setup PrimeTime For Lab 2

1. Invoke PrimeTime from the **lab1-6_reports** workshop lab Unix directory and restore the PrimeTime session using the **orca_savesession** directory.
2. Execute the following command to generate a timing report for **PCI_CLK**:

```
pt_shell> report_timing -group PCI_CLK
```

Question 1. Does this timing path meet or violate timing?

.....

Task 2. Analyze a Timing Report for Setup

1. Answer the following questions regarding the report generated in the last task.

Question 2. What type of timing path is this - internal flip-flop to flip-flop, input, or output timing path?

.....

Question 3. List the user specified constraints in this timing report.

.....

Task 3. Analyze a Timing Report for Hold

1. Generate a timing report for hold time for the same clock group **PCI_CLK**.

```
pt_shell> report_timing -group PCI_CLK -delay min
```

Question 4. What type of timing path is this - internal flip-flop to flip-flop, input, or output timing path?

.....

Question 5. How many cells are on the data path of this timing path?

.....

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Question 6. The cell delay used for the clock pin (**CP**) to **Q** pin of the start point flip-flop is for a rise transition. Offer one possible reason why this results in a worse slack for hold than using the faster fall delay through this flip-flop?

.....

In the next step, you will continue to explore and confirm your answer for the above question.

Question 7. What additional information do you need to confirm your answer for the above question?

.....

2. Generate another timing report for the same timing path for hold time but with a fall transition at the end point (instead of a rise transition).

Use copy and paste to avoid mistyping the end point and start point pin names.

Use the job aid labeled "timing reports" to find the appropriate switches for report_timing.

Question 8. Which lines in this report did you use to confirm that the correct path has been reported?

.....

Question 9. Was the guess correct – the faster fall delays results in a faster data arrival time but a smaller hold time requirement and thus a better slack?

.....

3. Quit PrimeTime.

This completes Lab 2. Return to lecture.

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Answers / Solutions

Question 1. Does this timing path meet or violate timing?

It violates timing with a slack of -1.157ns.

Question 2. What type of timing path is this - internal flip-flop to flip-flop, input, or output timing path?

This is an output timing path ending at the output port named **pad[1]**.

Question 3. List the user specified constraints in this timing report.

The clock period for **PCI_CLK** (15ns). The clock is propagated, so the network delay is calculated – network delay is NOT a constraint. The output external delay (which comes from an output delay constraint of 4ns).

Question 4. What type of timing path is this - internal flip-flop to flip-flop, input, or output timing path?

This is an internal timing path. The end point looks like a flip-flop, but is one of many timing arcs in a RAM timing model.

Question 5. How many cells are on the data path of this timing path?

There is only one cell on this data path, the start point flip-flop. The timing path consists of a start point flip-flop tied directly to the end point flip-flop.

Question 6. The cell delay used for the clock pin (**CP**) to **Q** pin of the start point flip-flop is a rise delay. Offer one reason why this would result in a worse slack for hold than using a fall delay through this flip-flop?

Typically, fall delays are faster than rise delays and would offer a worse slack for hold! The one exception is if the fall delay at the data pin of the end point flip-flop resulted in a smaller library hold time requirement. This is what occurs in this case.

Question 7. What additional information do you need to confirm your answer for the above question?

Generate another timing report where the data arrival time is calculated with fall transition at the end point and compare the two reports. In this way you can confirm that the library hold time is in fact smaller with a falling transition at the data pin of the end point flip-flop and thus the resulting slack better. You will explore this in the next lab step.

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Question 8. Which lines in this report did you use to confirm that the correct path has been reported?

Note: The backslash in the command below is a line continuation character.

```
pt_shell> report_timing -delay min_fall \
                    -to I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM/A1[1] \
                    -from I_ORCA_TOP/I_PCI_READ_FIFO/count_int_reg[1]1/CP

Startpoint: I_ORCA_TOP/I_PCI_READ_FIFO/count_int_reg[1]1
            (rising edge-triggered flip-flop clocked by PCI_CLK)
Endpoint: I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM
            (rising edge-triggered flip-flop clocked by PCI_CLK)
Path Group: PCI_CLK
Path Type: min
Min Data Paths Derating Factor  : 0.900
Min Clock Paths Derating Factor : 0.900

Point                                                    Incr       Path
-----------------------------------------------------------------------------
clock PCI_CLK (rise edge)                               0.000      0.000
clock network delay (propagated)                        0.779      0.779
I_ORCA_TOP/I_PCI_READ_FIFO/count_int_reg[1]1/CP (sdcrq1)
                                                        0.000      0.779 r
I_ORCA_TOP/I_PCI_READ_FIFO/count_int_reg[1]1/Q (sdcrq1)
                                                        0.344 &    1.123 f
I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM/A1[1] (ram32x32)
                                                        0.016 &    1.139 f
data arrival time                                                  1.139

clock PCI_CLK (rise edge)                               0.000      0.000
clock network delay (propagated)                        1.088      1.088
clock reconvergence pessimism                          -0.274      0.814
I_ORCA_TOP/I_PCI_READ_FIFO/PCI_RFIFO_RAM/CE1 (ram32x32)            0.814 r
library hold time                                       0.211 *    1.025
data required time                                                 1.025
-----------------------------------------------------------------------------
data required time                                                 1.025
data arrival time                                                 -1.139
-----------------------------------------------------------------------------
slack (MET)                                                        0.114
```

Question 9. Was the guess correct – the faster fall delays results in a faster data arrival time but a smaller hold time requirement and thus a better slack?

Yes! The data arrival time is faster (1.139ns versus 1.150ns) but the hold time requirement is smaller (0.211ns versus 0.411ns) thus the slack is better than the original timing report. Recall that hold time (and setup time) are a function of the transition at the data pin of the flip-flop.

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3 Generating Timing Paths of Interest

Learning Objectives

After completing this lab, you should be able to:

- Apply the correct switches to control which and how many paths are reported for debugging or performing design specific analysis
- Apply supporting commands that can be used to focus timing reports to, from or through specific pins

Lab Duration: 30 minutes

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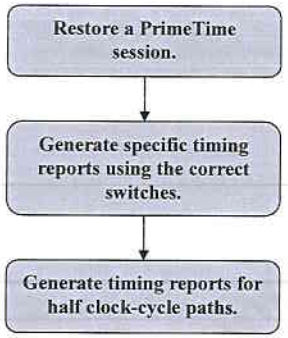
Overview



Relevant Files and Directories

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`lab1-6_reports/`	Current working directory
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`.synopsys_pt.setup`	PT setup file

Answers & Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question or would like to confirm your results, check the back portion of this lab.

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Instructions

Task 1. Setup PrimeTime For Lab 3

1. Invoke PrimeTime from the **lab1-6_reports** Unix lab directory and restore the PrimeTime session using the **orca_savesession** Unix directory.
2. Use help to list all of the available switches for **report_timing**.

Question 1. Which switch is useful for generating timing reports for paths with violations (i.e. with a slack less than zero)?

.....

Task 2. Apply the Correct Timing Report Switches

1. Turn page mode on.
2. Answer the following questions by experimenting and exploring in PrimeTime.

Use the job aid labeled "timing reports" for help identifying the appropriate commands and switches.

Question 2. Write the command to generate a single timing report for each path group for setup.

.....

Question 3. Write the command to generate a single timing report for setup for each path group which has a violation.

.....

Question 4. What are the names of the two path groups that have violating timing paths in **ORCA** (the answer will come from the result of the previous question)?

.....

Question 5. Write the command to generate a timing report with the worst slack for setup to any output port constrained by the clock **PCI_CLK**.

.....

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Question 6. There are a few latches in **ORCA**; write the command to identify the data pins of these latches.

.....

Question 7. Write the command to generate a timing report for hold to the **D** pin of the **latched_clk_en_reg** latches.

.....

Task 3. Identify Half-Clock Cycle Paths

The clock **SDRAM_CLK** constrains many half clock cycle paths in **ORCA** (i.e. it constrains paths from a falling edge triggered flip-flop to a rising edge triggered flip-flop and vice versa).

These paths must be carefully monitored for various reasons (e.g. the duty cycle of **SDRAM_CLK** is not yet well defined or for analysis of the clock skew).

1. Execute the following command to report the clock period for **SDRAM_CLK** and use this information to answer the following questions:

```
pt_shell> report_clock SDRAM_CLK
```

Question 8. Given that the first number under the waveform column is the first rising edge for the clock **SDRAM_CLK** and the second number is the falling edge – what duty cycle has been defined for this clock?

.....

Question 9. Describe the specific clock edges that will be used in a timing report for setup for a timing path constrained by the rising edge of **SDRAM_CLK** to the falling edge of **SDRAM_CLK**.

.....

Question 10. For this same timing path, describe the specific clock edges that will be used in a timing report for hold timing checks.

.....

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2. Confirm the information in the following table by generating the appropriate timing reports for the half clock cycle timing paths constrained by the clock **SDRAM_CLK**.

Launch clock edge	Capture clock edge	Worst Setup Slack	Launch clock edge	Capture clock edge	Worst Hold Slack
Rise 0ns	Fall 3.75ns	0.680ns	Rise 7.5ns	Fall 3.75ns	3.558ns
Fall 3.75ns	Rise 7.50ns	0.635ns	Fall 3.75ns	Rise 0ns	3.514ns

Question 11. Which switch is useful for generating the worst 10 timing reports for each of these half clock cycle timing paths?

.....

Question 12. Why does PrimeTime report "no constrained paths?" (hint – the options PrimeTime is using are shown immediately following the `report_timing` command)

.....

Question 13. What additional option must you use to report the worst 10 timing paths?

.....

3. Quit PrimeTime.

This completes Lab 3. Return to lecture.

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Answers / Solutions

Question 1. Which switch is useful for generating timing reports for paths with violations (i.e. with a slack less than zero)?

The `-slack_lesser_than 0` would report only a violating path if present. Consult man page for details.

```
pt_shell> report_timing -help
```

Question 2. Write the command to generate a single timing report for each path group for setup.

```
pt_shell> report_timing -group [get_path_group *]
```

Question 3. Write the command to generate a single timing report for setup for each path group which has a violation.

```
pt_shell> report_timing -group [get_path_group *]
-slack_lesser_than 0
# When using PrimeTime interactively - abbreviate
# command names or switches by typing enough letters to
# distinguish from other commands or switches - or,
# better yet, use command expansion by pressing tab
pt_shell> report_timing -slack_less 0
```

Question 4. What are the names of the two path groups that have violating timing paths in **ORCA** (the answer will come from the result of the previous question?

The two path groups are **PCI_CLK** and **SYS_CLK**.

Question 5. Write the command to generate a timing report with the worst slack for setup to any output port constrained by the clock **PCI_CLK**.

```
pt_shell> help all_*
pt_shell> all_outputs -help
pt_shell> report_timing -to [all_outputs -clock PCI_CLK]
# Or, another way to do the same thing
pt_shell> report_timing -to [all_outputs] -group PCI_CLK
```


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Question 6. There are a few latches in **ORCA**; write the command to identify the data pins of these latches.

```
pt_shell> all_registers -level_sensitive -data_pins
```

Question 7. Write the command to generate a timing report for hold to the **D** pin of the **latched_clk_en_reg** latches.

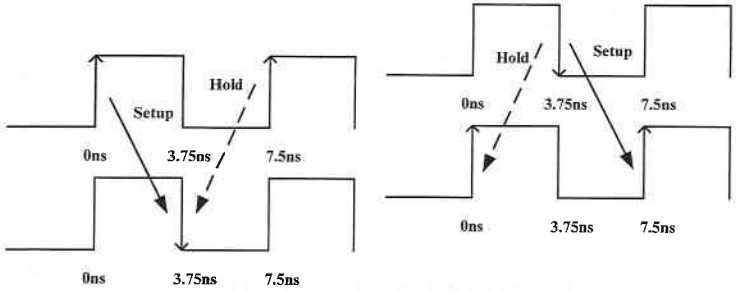
```
# Use copy and paste to avoid mistyping the long end point pin name
pt_shell> report_timing -delay min \
            -to I_ORCA_TOP/I_BLENDER*/latched_clk_en_reg/D
```

Question 8. Given that the first number under the waveform column is the first rising edge for the clock **SDRAM_CLK** and the second number is the falling edge – what duty cycle has been defined for this clock?

The rising edge of **SDRAM_CLK** is at 0ns, the falling edge at 3.75ns and the period is 7.50ns. The duty cycle is 50%.

Question 9. Describe the specific clock edges that will be used in a timing report for setup for a timing path constrained by the rising edge of **SDRAM_CLK** to the falling edge of **SDRAM_CLK**.

Use the following clock waveform for this and the next question. The clock edges will be 0ns to 3.75ns.



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Question 10. For this same timing path, describe the specific clock edges that will be used in a timing report for hold timing checks.

The clock edges will be 7.5ns to 3.75ns.

```
# Commands for the final task
# The backslash is a line continuation character
# The switch -delay min_max will generate one report for setup and
# one for hold
report_timing -rise_from [get_clocks SDRAM_CLK] \
            -fall_to [get_clocks SDRAM_CLK] -delay_type min_max
report_timing -fall_from [get_clocks SDRAM_CLK] \
 -rise_to [get_clocks SDRAM_CLK] -delay_type min_max
```

Question 11. Which switch is useful for generating the worst 10 timing reports for each of these half clock cycle timing paths?

The switch **-max_paths 10** to the above command.

Question 12. Why does PrimeTime report "no constrained paths?"

The **-max_paths** option implicitly sets another option: **slack_lesser_than 0**, which, because slack is positive, results in no constrained paths.

Question 13. What additional option must you use to report the worst 10 timing paths?

Since there are no violating paths that could be reported in this case, add the option: **slack_less_than 100**

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5 Summary Reports

Learning Objectives

After completing this lab, you should be able to:

- Generate summary reports for the violations in ORCA

Lab Duration: 15 minutes

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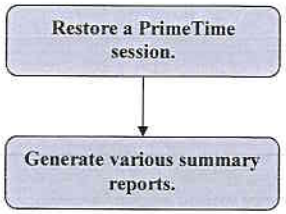
Overview



Relevant Files and Directories

All files for this lab are located in the **`lab1-6_reports`** directory under your home directory.

`lab1-6_reports/`	Current working directory
`orca_savesession/`	Session to restore for labs
`.synopsys_pt.setup`	PT setup file

Answers & Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question or would like to confirm your results, check the back portion of this lab.

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Instructions

Task 1. Setup PrimeTime For Lab 5

1. Invoke PrimeTime from the `lab1-6_reports` workshop lab directory.

   Restore the PrimeTime session using the `orca_savesession` directory.

2. Find the variable that controls the significant digits for many reports and modify this variable to 4 significant digits.

   Question 1. What was this variable set to before you modified it to 4?

   .....

Task 2. Generate Summary Reports

From lab 1, you know that there are 13 setup violations in **ORCA**.

1. Answer the following questions by generating the appropriate summary reports:

   Question 2. Identify the top five setup violations with the worst slack. The required details are the endpoint names and the slack.

   .....

   Question 3. List the 2 clock domains that have violating setup timing paths, and the 5 clock domains that have violating hold timing paths (**ORCA** has 6 clock domains in total).

   .....

   Question 4. Identify how many hold violations are on input paths, how many on output paths, and how many are register-to-register violations.

   .....

2. Generate a report for the worst slack for setup to each bit of a 16-bit bus ending at the output ports `sd_DQ[0]` to `sd_DQ[15]` (the output ports are all constrained by a single clock, `SD_DDR_CLK`).

   Question 5. List the end point with the largest margin (the best slack).

   .....

3. Generate a high-level overview of the quality of the design

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Question 6. Which clock group has the highest number of violating paths?

.....

4. Quit PrimeTime.

This completes Lab 5. End of Day-1.

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Answers / Solutions

Question 1. What was this variable set to before you modified it to 4?

```
pt_shell> printvar *sig*
→ report_default_significant_digits = "3"
pt_shell> set report_default_significant_digits 4
```

Question 2. Identify the top five setup violations with the worst slack. The details that are required are the endpoint names and the slack.

The following command will list all setup violations sorted by slack. Use page mode to quit from the long report because the only information desired are the top 5 violations.

```
# No need to type the entire command name!
pt_shell> report_analysis -status violated -check setup

Constrained                                Related          Check
Pin                                        Pin      Clock   Type    Slack
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/D      CP(rise) SYS_CLK setup  -0.9872
I_ORCA_TOP/I_BLENDER/s4_op1_reg[31]/D      CP(rise) SYS_CLK setup  -0.8410
I_ORCA_TOP/I_BLENDER/s4_op2_reg[30]/D      CP(rise) SYS_CLK setup  -0.8305
I_ORCA_TOP/I_BLENDER/s4_op1_reg[15]/D      CP(rise) SYS_CLK setup  -0.6918
I_ORCA_TOP/I_BLENDER/s4_op1_reg[30]/D      CP(rise) SYS_CLK setup  -0.6843
```

Question 3. List the 2 clock domains that have violating setup timing paths, and the 5 clock domains that have violating hold timing paths (**ORCA** has 6 clock domains in total). (the following answer just shows the path group headers – the names of the endpoints have been removed to conserve space.

```
pt_shell> report_constraint -all

max_delay/setup ('PCI_CLK' group)
max_delay/setup ('SYS_CLK' group)

min_delay/hold ('PCI_CLK' group)
min_delay/hold ('SDRAM_CLK' group)
min_delay/hold ('SD_DDR_CLK' group)
min_delay/hold ('SYS_2x_CLK' group)
min_delay/hold ('SYS_CLK' group)
```


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Question 4. Identify how many hold violations are on input paths, how many on output paths, and how many are register-to-register violations.

```
pt_shell> report_global_timing

Hold violations
-----------------------------------------------
        Total    reg->reg in->reg reg->out in->out
-----------------------------------------------
WNS     -0.4375  -0.1420 -0.2363 -0.1281  -0.4375
TNS    -12.6317  -3.5616 -2.5768 -0.5954  -5.8979
NUM       100        59      18       7       16
```


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Question 5. List the end point with the largest margin (the best slack).

The output port `sd_DQ[0]` has the largest margin at 1.5994ns.

Generally, the following command will only generate a single report for every end point because nworst is, by default, 1 and there is only a single clock constraining every output port. However, because increasing the value of `max_paths` causes an implicit `slack_lesser` 0 to be used, and because all the slacks to this endpoint are positive, PrimeTime will not report any paths unless we change the value of `slack_lesser` to a large positive number – in this case, 100.

```
pt_shell> report_timing -path end -max 16 -slack_lesser 100 -to sd_DQ*
****************************************
Report : timing
        -path_type end
        -delay_type max
        -slack_lesser_than 100.0000
        -max_paths 16
        -sort_by slack
Design : ORCA
Version: J-2014.06
Date   : Wed Jul 23 12:01:51 2014
****************************************
Endpoint              Path Delay   Path Required    CRP      Slack

sd_DQ[14] (inout)      8.4133 f       9.6649       0.2418   1.4933
sd_DQ[6] (inout)       8.4133 f       9.6649       0.2418   1.4934
sd_DQ[5] (inout)       8.4133 f       9.6649       0.2418   1.4934
sd_DQ[4] (inout)       8.4133 f       9.6649       0.2418   1.4934
sd_DQ[3] (inout)       8.4133 f       9.6649       0.2418   1.4934
sd_DQ[10] (inout)      8.4112 f       9.6649       0.2418   1.4955
sd_DQ[15] (inout)      8.4111 f       9.6649       0.2418   1.4955
sd_DQ[13] (inout)      8.4111 f       9.6649       0.2418   1.4955
sd_DQ[12] (inout)      8.4111 f       9.6649       0.2418   1.4955
sd_DQ[9] (inout)       8.4111 f       9.6649       0.2418   1.4955
sd_DQ[11] (inout)      8.4106 f       9.6649       0.2418   1.4960
sd_DQ[2] (inout)       8.4054 f       9.6649       0.2418   1.5013
sd_DQ[1] (inout)       8.4054 f       9.6649       0.2418   1.5013
sd_DQ[8] (inout)       8.4032 f       9.6649       0.2418   1.5035
sd_DQ[7] (inout)       8.4032 f       9.6649       0.2418   1.5035
sd_DQ[0] (inout)       8.0654 f*      9.6649       0.0000   1.5994
```

Question 6. Which clock group has the highest number of violating paths?

```
report_qor -only_violated
```

SDRAM_CLK has 30 violating paths

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6 Constraints in a Timing Report

Learning Objectives

After completing this lab, you should be able to:

- Identify and interpret constraints in a timing report
  - Clock constraints
  - Interface constraints

Lab Duration: 30 minutes

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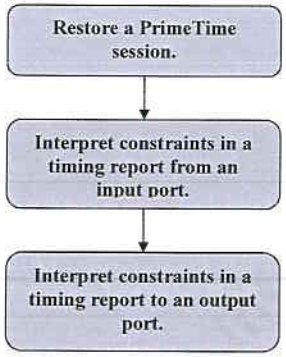
Overview



Relevant Files and Directories

All files for this lab are located in the `lab1-6_reports` directory under your home directory.

`lab1-6_reports/`	Current working directory
`orca_savesession/`	Session to restore for labs
`.synopsys_pt.setup`	PT setup file

Answers & Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question or would like to confirm your results, check the back portion of this lab.

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Instructions

Task 1. Setup PrimeTime For Lab 6

1. Invoke PrimeTime from the **lab1-6_reports** workshop lab Unix directory and restore the PrimeTime session using the **orca_savesession** directory.

   Use the job aid labeled "timing reports" if you have forgotten the command.

2. Generate a summary report for all violations in **ORCA**.

   Use the job aid labeled " timing reports" if you have forgotten the command.

   Question 1. How many output delay constraints are there for setup and for hold and are these constraints met or violated?

   .....

Task 2. Analyze a Timing Report From an Input Port

1. Generate a report for the input delay constraints applied to the port **pad[0]** .

```
pt_shell> report_port -input_delay pad[0]
```

   Question 2. What are the min and max arrival times to **pad[0]** ?

   .....

   Question 3. What is the name of the external start point clock constraining **pad[0]** ?

   .....

2. Generate a timing report for setup starting at the port **pad[0]** .

   Answer the following questions using this report.

   Use your job aid labeled "timing reports" for help recalling the appropriate switch for report_timing.

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Question 4. Which lines in the timing report did you use to ensure the reported path starts at the port **pad[0]** and is for setup?

.....

Question 5. List all user specified constraints in this timing report.

.....

Question 6. Where must the clock latency be included for the start point clock **PCI_CLK**?

.....

Question 7. Describe the direction of the port **pad[0]** (i.e. is it an input, output or inout port).

.....

Question 8. Describe the end point of this timing path (i.e. is it an output port or an internal flip-flop).

.....

3. Generate a new report from the same port **pad[0]** for setup, which also shows the details of the calculated clock network delay.

   Use the job aid labeled "timing reports" for help recalling the appropriate switch for report_timing. Remember to take advantage of history commands.

   Question 9. How large is the clock source latency versus the clock network latency for the end point clock **PCI_CLK**?

   .....

   Question 10. Where has the clock **PCI_CLK** been defined (the clock definition point)?

   .....

4. Generate a report starting at the port **pad[0]** for hold time.

   Question 11. Does the value of the input external delay constraint match your expectations?

   .....

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Task 3. Analyze a Timing Report to an Output Port

1. Generate a report for the output delay constraints applied to the port **pad[0]**.

   Use help in PrimeTime to determine the correct command and switch to do this – using the previous task as an example.

   Question 12. What are the min and max output delay constraints for this port?

   .....

   Question 13. How will the negative min output delay constraint be applied to this port (i.e. will it impose a positive or negative hold requirement)?

   .....

   Question 14. What is the name of the external end point clock constraining this port?

   .....

2. Generate a "short" timing report ending at the port **pad[0]** for hold time.

   Question 15. Describe the start point of this timing path (i.e. is it an input port or an internal flip-flop).

   .....

   Question 16. Does the path group for this timing path match your expectations?

   .....

   Question 17. Does the "data required time" match your expectations?

   .....

3. Optionally, apply the following constraint which will impose a positive output delay constraint for hold on **pad[0]** and then re-execute the steps in this task to see the affect.

```
pt_shell> set_output_delay -min 1.0 -clock PCI_CLK pad[0]
```

   Quit PrimeTime.

   This completes Lab6. Lab6a is optional. Return to lecture.

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Answers / Solutions

```
pt_shell> restore_session orca_savesession
pt_shell> report_analysis_coverage
```

Type of Check	Total	Met	Violated	Untested
setup	9629	3575 (37%)	13 (0%)	6041 (63%)
hold	9629	3517 (37%)	71 (1%)	6041 (63%)
recovery	1316	1210 (92%)	0 (0%)	106 (8%)
removal	1316	1204 (91%)	6 (0%)	106 (8%)
min_period	20	20 (100%)	0 (0%)	0 (0%)
min_pulse_width	7273	5957 (82%)	0 (0%)	1316 (18%)
out_setup	75	66 (88%)	9 (12%)	0 (0%)
out_hold	75	36 (48%)	39 (52%)	0 (0%)
All Checks	29333	15585 (53%)	138 (0%)	13610 (46%)

Question 1. How many output delay constraints are there for setup and for hold and are these constraints met or violated?

From **report_analysis_coverage**, there are 75 output delay constraints for both setup and hold; 9 output delay constraints for setup are violated; 39 output delay constraints for hold are violated. Remember to verify that all output ports are constrained for both setup as well as for hold.

Question 2. What are the min and max arrival times to **pad[0]**?

The min and max arrival times are 2ns and 8ns respectively (with the same constraint for both rise and fall data transitions at the port pad[0]).

Question 3. What is the name of the external start point clock constraining **pad[0]**?

The name of the clock is **PCI_CLK**.

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Question 4. Which lines in the timing report did you use to ensure the reported path starts at the port **pad[0]** and is for setup?

```
pt_shell> report_timing -from pad[0]
Startpoint: pad[0] (input port clocked by PCI_CLK)
Endpoint: I_ORCA_TOP/I_PCI_CORE/d_out_i_bus_reg[0]
            (rising edge-triggered flip-flop clocked by PCI_CLK)
Path Group: PCI_CLK
Path Type: max
```

Question 5. List all user specified constraints involved in this timing report.

The clock period is a constraint. The clock **PCI_CLK** is propagated (not ideal). The input external delay (which comes from an input delay constraint).

Question 6. Where must the clock latency be included for the start point clock **PCI_CLK**?

The clock network delay is zero. Therefore, the only other place to represent the external clock latency is as a part of the input delay constraint (i.e. the input external delay). The appropriate way to model this is to use the switches **-network_latency_included** and **-source_latency_included** for **set_input_delay.**

Question 7. Describe the direction of the port **pad[0]** (i.e. is it an input, output or inout port).

The port **pad[0]** is an inout port; therefore, it is both a timing path start point as well as a timing path end point!

```
Point                                          Incr       Path
-----------------------------------------------------------------
clock PCI_CLK (rise edge)                      0.000      0.000
clock network delay (propagated)               0.000      0.000
input external delay                           8.000      8.000 r
pad[0] (inout)                                 0.000      8.000 r
```

Question 8. Describe the end point of this timing path (i.e. is it an output port or an internal flip-flop).

The end point is a rising-edge triggered flip-flop clocked by **PCI_CLK** (it is actually a timing model that looks like a flip-flop with setup and hold timing checks).

!report_t

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Question 9. How large is the clock source latency versus the clock network latency for the end point clock **PCI_CLK**?

Shown below is only the data required time section of the timing report. The source latency is 0ns. The clock network latency is 0.768ns (15.768 – 15.000).

```
pt_shell> !rep -path full_clock
. . .

clock PCI_CLK (rise edge)                                  15.000      15.000
clock source latency                                        0.000      15.000
pclk (in)                                                   0.000      15.000 r
pclk_iopad/CIN (pc3d01)                                     0.728 H    15.728 r
I_CLOCK_GEN/I_PLL_PCI/CLK (PLL)                            -1.220 H    14.508 r
I_CLOCK_GEN/bufbdfG1B1I1_1/Z (bufbdf)                       0.184 &    14.692 r
I_CLOCK_GEN/U21/Z (mx02d2)                                  0.165 &    14.857 r
I_CLOCK_GEN/bufbdfG2B1I1_2/Z (bufbdf)                       0.153 &    15.010 r
I_CLOCK_GEN/U17/ZN (invbdk)                                 0.076 &    15.086 f
I_CLOCK_GEN/U14/ZN (invbdk)                                 0.059 &    15.145 r
I_CLK_SOURCE_PCLK/Z (bufbdk)                                0.168 &    15.313 r
invbd7G5B1I2/ZN (invbd7)                                    0.108 &    15.421 f
I_ORCA_TOP/invbdkG5B2I3_1/ZN (invbdk)                       0.120 &    15.541 r
I_ORCA_TOP/I_PCI_CORE/buffd7G5B3I32/Z (buffd7)              0.140 &    15.680 r
I_ORCA_TOP/I_PCI_CORE/d_out_i_bus_reg[0]/CP (sdcrq1)
                                                            0.088 &    15.768 r
library setup time                                         -0.153      15.615
data required time                                                     15.615
```

Question 10. Where has the clock **PCI_CLK** been defined (the clock definition point)?

The clock **PCI_CLK** is defined at the input port **pclk**. The clock definition point separates the clock source latency from the clock network latency.

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Question 11. Does the value of the input external delay constraint match your expectations?

Yes. From **report_port** above, the input external delay should be 2ns with respect to the rising edge of **PCI_CLK**. This is confirmed in the timing report below.

pt_shell> **report_timing -delay min -from pad[0]**

```
Startpoint: pad[0] (input port clocked by PCI_CLK)
Endpoint: I_ORCA_TOP/I_PCI_CORE/d_out_i_bus_reg[0]
            (rising edge-triggered flip-flop clocked by PCI_CLK)
Path Group: PCI_CLK
Path Type: min
Min Data Paths Derating Factor   : 0.900
Min Clock Paths Derating Factor : 0.900

Point                                                    Incr       Path
-----------------------------------------------------------------------------
clock PCI_CLK (rise edge)                                0.000      0.000
clock network delay (propagated)                         0.000      0.000
input external delay                                     2.000      2.000 r
pad[0] (inout)                                           0.000      2.000 r
pad_iopad_0/PAD (pc3b03)                                 0.044      2.044 r
pad_iopad_0/CIN (pc3b03)                                 0.662 &    2.706 r
```

Question 12. What are the min/max output delay constraints for this port?

```
pt_shell> report_port -help

Usage:
 report_port            # Report port info
   [-verbose]               (Show all port info)
   [-design_rule]           (Only port design rule info)
   [-drive]                 (Only port drive info)
   [-input_delay]           (Only port input delay info)
   [-output_delay]          (Only port output delay info)
   [-wire_load]             (Only port wire load info)
   [-nosplit]               (Don't split lines if column overflows)
   [port_list]              (List of ports)

pt_shell> report_port -output_delay pad[0]

                    Output Delay
                  Min            Max        Related Related
 Output Port    Rise   Fall   Rise   Fall   Clock   Pin
-------------------------------------------------------------
pad[0]        -1.00  -1.00   4.00   4.00  PCI_CLK   --
```


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Question 13. How will the negative min output delay constraint be applied to this port (i.e. will it impose a positive or negative hold requirement)?

In lecture, it was stated that a negative hold output delay constraint will impose a positive hold requirement.

Question 14. What is the name of the external end point clock constraining this port?

The port **pad[0]** is constrained with respect to **PCI_CLK**.

Question 15. Describe the start point of this timing path.

The start point of the timing path is an internal flip-flop.

```
pt_shell> report_timing -delay min -to pad[0]

Startpoint: I_ORCA_TOP/I_PCI_CORE/pad_out_buf_reg[0]
            (rising edge-triggered flip-flop clocked
by PCI_CLK)
Endpoint: pad[0] (output port clocked by PCI_CLK)
```

Question 16. Does the path group for this timing path match your expectations?

Yes. The path group is **PCI_CLK** which is the same as the external capture clock name.

Question 17. Does the "data required time" match your expectations?

Yes. The capture clock edge is zero, which is appropriate for hold. The hold requirement of 1ns is positive, and the propagated clock network delay is 0ns. The data required section of the timing report is shown below.

```
clock PCI_CLK (rise edge)                    0.000      0.000
clock network delay (propagated)             0.000      0.000
output external delay                        1.000      1.000
data required time                                      1.000
```


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```
# Answers for optional step
pt_shell> report_port -output_delay pad[0]
                    Output Delay
                   Min              Max        Related Related
 Output Port     Rise    Fall    Rise    Fall   Clock   Pin
-----------------------------------------------------------
  pad[0]         1.00    1.00    4.00    4.00  PCI_CLK   --

pt_shell> report_timing -to pad[0] -delay min -path short
* * *
  Point                                                   Incr       Path
  ---------------------------------------------------------------------------
  clock PCI_CLK (rise edge)                               0.000      0.000
  clock network delay (propagated)                        0.772      0.772
  I_ORCA_TOP/I_PCI_CORE/pad_out_buf_reg[0]/CP (sdcrq1)
                                                          0.000      0.772 r
  I_ORCA_TOP/I_PCI_CORE/pad_out_buf_reg[0]/Q (sdcrq1)     0.350 &    1.122 r
  pad_out_buf_reg_0_ASTttcInst778/Z (bufbd4)              0.199 &    1.321 r
  pad_iopad_0/PAD (pc3b03)                                2.081 H    3.402 r
  pad[0] (inout)                                          0.044      3.445 r
  data arrival time                                                  3.445

  clock PCI_CLK (rise edge)                               0.000      0.000
  clock network delay (propagated)                        0.000      0.000
  clock reconvergence pessimism                           0.000      0.000
  output external delay                                  -1.000     -1.000
  data required time                                                -1.000
  ---------------------------------------------------------------------------
  data required time                                                -1.000
  data arrival time                                                 -3.445
  ---------------------------------------------------------------------------
  slack (MET)                                                        4.445

#From the above report, you can see that using a positive hold constraint
#for an output delay INCREASES the positive slack.  This confirms that
#specifying a negative hold constraint for an output delay is correct.
```


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[6a] PrimeTime Objects (OPTIONAL Lab)

Learning Objectives

After completing this lab, you should be able to:

- Display objects using the '`all...`' and '`get...`' commands
- Traverse a design using the `-of_objects` option.
- Ease interactive typing by defining and dereferencing a variable.

Lab Duration: 15 minutes

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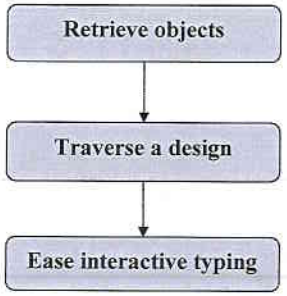
Overview



Relevant Files and Directories

All files for this lab are located in the `lab1-6_reports` directory under your home directory.

`lab1-6_reports/`	Current working directory
`orca_savesession/`	Session to restore for labs
`.synopsys_pt.setup`	PT setup file

Answers & Solutions

This lab guide contains answers and solutions to all questions. If you need some help with answering a question or would like to confirm your results, check the back portion of this lab.

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Instructions

Your goal is to gain familiarity with PrimeTime objects, with accessing those objects, and with seeing how objects are connected. In addition, you will create and dereference a variable.

Task 1. Set up PrimeTime for Lab 6a

1. Invoke PrimeTime from the **lab1-6_reports** workshop lab Unix directory and restore a saved session.

```
unix% cd lab1-6_reports
unix% pt_shell
pt_shell> restore_session orca_savesession
```

Task 2. Traverse a Design from Port to Clock Generator

This task shows '**scripting techniques**' to find out what is connected to a cell, or to a port, or to a net; that is, to traverse a design.

The PrimeTime GUI and the report... commands, covered later in this course, are alternative powerful **interactive** ways to traverse a design.

1. Display all the input ports.

```
pt_shell> all_inputs
```

2. Narrow the display to ports containing the string 'clk'

```
pt_shell> get_ports *clk*
```

3. Display nets connected to the port '**pclk**'.

```
get_nets -of_objects [get_ports pclk]
```


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4. Display cells connected to the net '**pclk**'.

```
get_cells -of_objects [get_nets pclk]
```

5. Look further downstream; display nets connected to the cell 'pclk_iopad'

```
get_nets -of_objects [get_cells pclk_iopad]
```

6. Look still further downstream; display cells connected to the net '**net_pclk**'

```
get_cells -of_objects [get_nets net_pclk]
```

7. Display pins of **I_CLOCK_GEN.**

```
get_pins -of_objects [get_cells I_CLOCK_GEN]
```

8. Set up an interactive shortcut by assigning a long name to a variable. In this case, cut and paste any long pin name and assign it to the variable 'a'.

```
set a I_CLOCK_GEN/o_sys_2x_clk
```

9. Use the variable you created.

```
get_cell -of $a
```

Question 1. If you know the name of a net, what objects can you access using the **-of_objects** option?

.....

Question 2. To get additional detailed information about any of the objects you retrieve with the '**get...**' commands, what ready-made commands would you try?

.....

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Task 3. Optional – A Little More Power --

These tasks are not necessary to succeed with any of the next modules in the course. However, they give you more power to explore PrimeTime objects and how they are connected in your design.

1. Look at the values of attributes on the clock object '**PCI_CLK**'.

```
get_attribute [get_clock PCI_CLK] period
report_attribute -application [get_clocks PCI_CLK]
report_attribute -application [get_clocks PCI_CLK] \
-attributes period
```

Question 3. Which command would you use interactively?

.....

Question 4. Which command can be nested or embedded within another command, making it useful in a script?

.....

2. Display the names and class of objects connected to a net.

```
set my_net [get_nets pclk]
query_objects -verbose [all_connected $my_net]
```

Question 5. What classes of objects are connected to the **pclk** net?

.....

This completes Lab 6a, that was optional. Return to lecture.

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Answers / Solutions

Question 1. If you know the name of a net, what objects can you access using the **-of_objects** option?

Cells, pins, and ports:

```
set my_net [get_nets pclk]
get_ports -of_objects $my_net
get_pins -of_objects $my_net
get_cells -of_objects $my_net
```

Question 2. To get additional detailed information about any of the objects you retrieve with the '**get...**' commands, what ready-made commands would you try?

report... commands. Do a '**help report***" to display a list of commands that present information for interactive use.

Question 3. Which command would you use interactively?

report_attribute is an interactive display command; **get_attribute** can be used interactively or in scripting.

Question 4. Which command can be nested or embedded within another command, making it useful in a script?

get_attribute – it returns a value.

Question 5. What classes of objects are connected to the pclk net?

Ports and pins.

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8 Constraining Multiple Clocks

Learning Objectives

After completing this lab, you should be able to:

- Apply the commands taught in lecture to gather information about the design clocks
- Use the GUI for another view of the design clocks and their relationships

Lab Duration: 60 minutes

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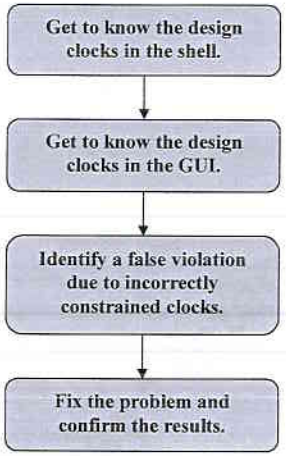
Overview



Relevant Files and Directories

All files for this lab are located in the **`lab8_clocks`** directory under your home directory.

`lab8_clocks/`	Current working directory
`orca_savesession/`	Saved ORCA session
`RUN.tcl`	Run script for ORCA
`scripts/`	
`orca_pt_variables.tcl`	Variable script with a typo
`.solutions/`	
`orca_pt_variables.tcl`	Fixed variable script for reference

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Instructions

Task 1. Get to Know the Design Clocks

1. Make sure your current directory is `lab8_clocks`
2. Invoke PrimeTime (pt_shell).

   Restore the session saved in `./orca_savesession`

   Take advantage of command and file name completion by typing a few letters and then using the tab key.

3. Use the commands taught in lecture to answer the following questions.

   Use the job aid labeled "Clocks and More" for help recalling the specific commands.

   Question 1. How many clocks are in this design and how many of these are generated?

   .....

   Question 2. Which input ports have defined, master clocks?

   .....

   Question 3. Which output ports have defined, outgoing clocks?

   .....

   Question 4. Are the clocks propagated or ideal?

   .....

   Question 5. Which 3 clock pairs have constrained timing paths?

   .....

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Task 2. Use the GUI to Report Clock Relationships

If your design has many clocks, the GUI may simplify the task of understanding how the clocks are related.

1. Start the GUI by executing the following command.

```
pt_shell> start_gui
```

   Note: The original pt_shell session is still running in the terminal window. You can keep the GUI open and use either the shell or the GUI interface as appropriate to the desired tasks.

2. Look at clock domain crossings: Open the "clock domain matrix" from the pull-down menu: **Clock → Clock Analyzer.**

   The *ClockAnalyzer* window that opens (expand if needed by clicking on the plus signs to the left of the clocks) should match the information from **`check_timing`** when reporting the clock crossings in the design. Mouse over the blocks in the matrix to see information on what type of false paths exist. It is sometimes easier to digest this information as a graphical matrix table in comparison to the text output from **`check_timing -override clock_crossing -verbose`**.

   The left part of the window lists each master clock and any generated clocks that are created from each master clock.

   Question 6. What is the master clock for **`SYS_2x_CLK`**?

   .....

   Question 7. **`SYS_2x_CLK`** is defined on which pin/port (its "source")? (note: you may have to drag the clock matrix out of the way, exposing more columns of information about the clocks)

   .....

   Question 8. The master clock for **`SYS_2x_CLK`** is defined on which pin/port?

   .....

3. Explore in more detail by displaying the clock schematic for SYS_2x_CLK: select the clock, then right mouse button->`Schematic of Selected Clocks`. Expand the fanin for the schematic for the MUX called I_CLOCK_GEN/U20 by double-clicking the input stubs, as shown in the

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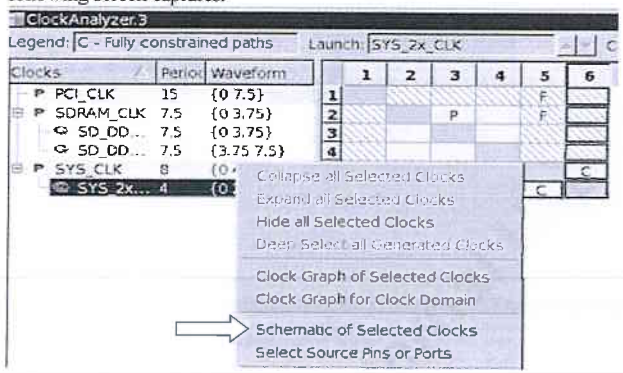
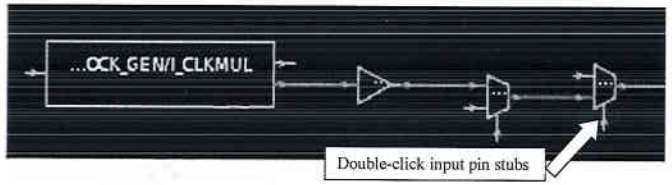
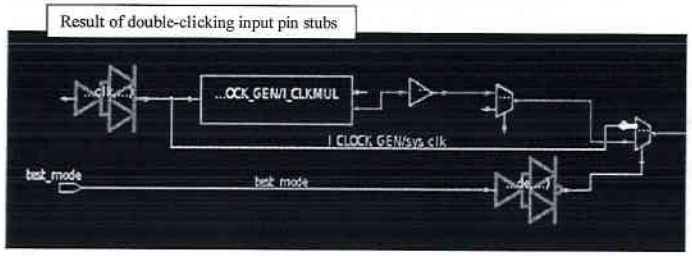
following screen captures:







Question 9. What port is connected to the select pin of the MUX I_CLOCK_GEN/U20?

.....

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Question 10. Does seeing the schematic give you insight into the clocking scheme for test?

.....

4. Explore clock relationships with the abstract clock graph: Close the schematic window, then, on the TopLevel window, select `Clock-> Clock Graph for All Clocks`. If necessary, display a toolbar next to the schematic by pressing the F8 key. Display various elements by checking the toolbar and pressing Apply.
5. Find a pair of muxed clocks: In the Abstract Clock Graph toolbar, select Mux and click Apply.
6. In the Abstract Clock Graph, find instance I_CLOCK_GEN/U10 of mx02d1.

Question 11. What clocks drive I_CLOCK_GEN/U10?

.....

7. From the clock graph window, 'zoom into' an interesting object by displaying a schematic for it: Select I_CLOCK_GEN/U10, then right-mouse-click and choose Path Schematic for Selected Paths.

Question 12. What port drives the select line to I_CLOCK_GEN/U10?

.....

8. Go back to the Abstract Clock Graph.

Question 13. From the abstract clock graph window, is it possible to open and display the same clock schematic for SYS_2x_CLK you displayed in the clock analyzer?

.....

9. Close the *Clock Analyzer* window by clicking on the small "**X**" in its upper right corner.
10. Close the *Clock Schematic* and *Clock Analyzer* windows by clicking on the small "**X**" in the upper right corner.

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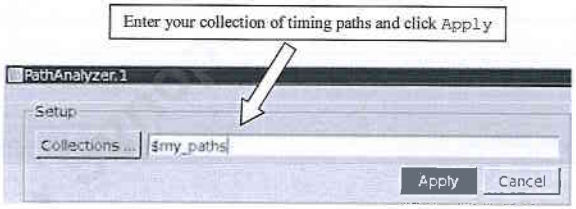
Task 3. Use the GUI to explore detail of timing paths

Investigate paths between launch and capture clocks – in this case, you will look at network latency for the launch and capture paths clocked by SYS_CLK.

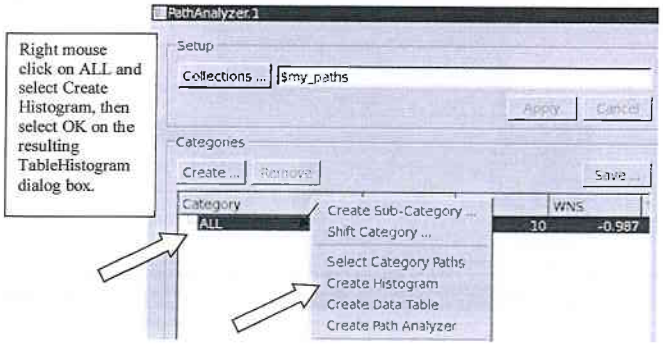
1. Make Primetime calculate the actual (propagated) latency before examining paths, by executing these commands in the shell, which remains open behind the GUI (this will take a minute or so to complete). Tell PrimeTime to save the arrival times for all pins (this is what you will examine). Then, define a collection of timing paths to examine.

```
set_propagated_clock [all_clocks]
set timing_save_pin_arrival_and_slack true
update_timing
set my_paths [get_timing_paths -max 10 -group SYS_CLK
-path full_clock_expanded]
```

2. Enter your collection of violating paths from the pull-down menu **Timing → New Path Analyzer**.



3. Bring up a histogram of your ten timing paths.



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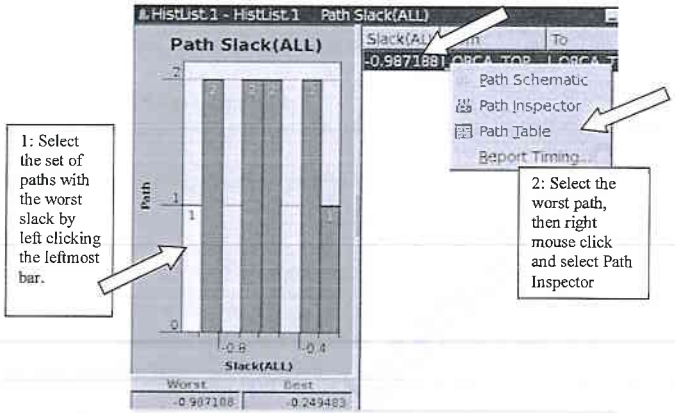
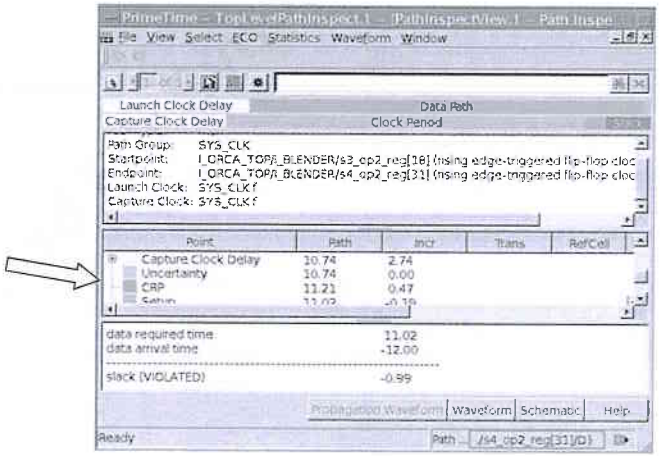
4. From the histogram, bring up the Path Inspector on a selected path.



5. In the Path Inspector, examine clock reconvergent pessimism: In the data required and data arrival section, scroll down until you find CRP. Then, scroll across until you find the percent of delay for the CRP.



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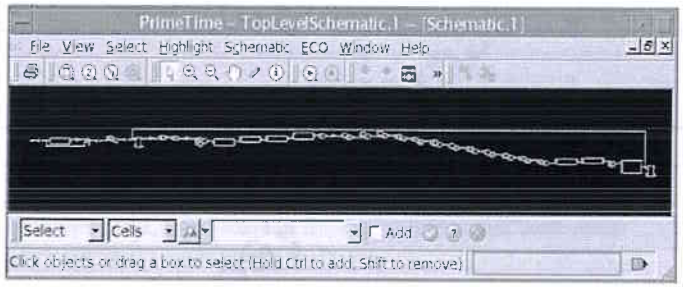
Question 14. What percent of the capture delay comes from CRP?

.....

Question 15. Is this percent representative of all designs?

.....

6. Look at a schematic of the path by clicking on the `Schematic` tab on the bottom of the path inspector window.



7. In the schematic window, find the CRP (clock reconvergent pessimism) point. This is the last pin before the launch and capture paths diverge.

Note: Mouse "gestures" or "strokes" are available for easier zooming: While pressing the middle mouse button drag the cursor vertically for 'zoom full'; Drag diagonally up across an object to zoom in, and down across an object to zoom out.

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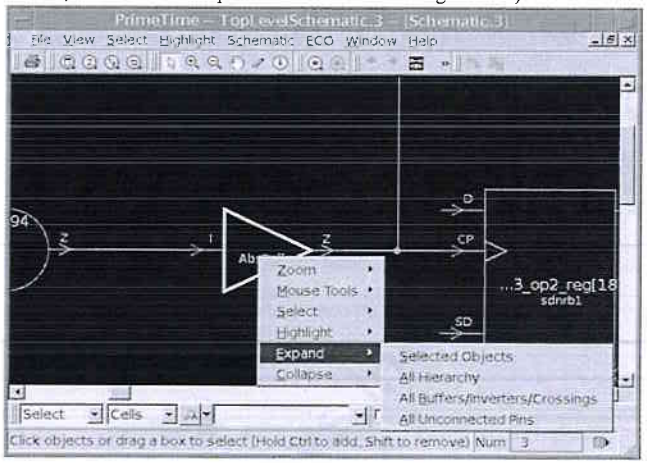
8. To see arrival times on this pin, you must first 'expand' the pin's buffer. (By default, PrimeTime 'collapses' buffer trees into a single buffer).



9. View the arrival times (and any other attributes of interest) by selecting the output pin of the buffer just before the register, then by selecting View->Property

Question 16. How wide is the arrival window for thee buffer output pin?

.....

Question 17. Does this match what we saw earlier in the data arrival data required section of the path inspector?

.....

10. Examine the path waveform: Click on the Waveform tab at the bottom of the Path Inspector window.

Question 18. What can you add to the wavforms by clicking the right mouse button in the waveform window?

.....

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11. Close the GUI while keeping the original pt_shell session going in the terminal window:

```
File → Close GUI (in the main GUI window)
Or
pt_shell> stop_gui (in the pt_shell window)
```

12. Exit PrimeTime.

Task 4. Report a False Violation

1. Bring up PrimeTime and restore the saved session **orca_savesession**.
2. Determine the number and type of timing violations in **ORCA**:

```
report_analysis_coverage
```

Question 19. How many, and what kind of violations does **ORCA** have?

.....

3. Generate a "short" timing report for the worst slack for an **out_setup** timing check.

Question 20. How will you identify the endpoint port which has the worst slack for **out_setup** (use the job aid labeled "Timing Reports" for help recalling the two appropriate switches)?

.....

Question 21. Which clocks (launch and capture) are involved in this violation?

.....

From task 1, you know that SD_DDR_CLK is a generated clock defined at an output port. The purpose of defining outgoing clocks is that PrimeTime calculates source latency for this clock and includes this latency as part of the data required time.

4. Look at the data required time section of the timing report from the last step and notice that no clock latency is reported.

   Confirm this with the following command:

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```
# This report will return nothing as PrimeTime has not
# calculated source latency for SD_DDR_CLK
pt_shell> report_clock -skew SD_DDR_CLK
```

Question 22. Why has PrimeTime not calculated source latency for the outgoing clock **SD_DDR_CLK**?

.....

After speaking with the designer, it turns out there was a miscommunication. The designer was expecting you to turn on a variable that will propagate all clocks!

5. There is a variable that can be used to make all clocks propagated. Use the Tcl procedure aa to help you identify the appropriate variable:

```
aa propagate
```

Question 23. What is the name of this variable?

.....

Question 24. Using a man page, explain what this variable will do?

.....

6. Use the man page for check_timing to find the name of the additional check that will flag all ideal clocks.

   The following command opens the man page in a pop-up window with a scroll bar that simplifies viewing long reports.

```
pt_shell> vman check_timing
```

The above command is an alias created in the .synopsys_pt.setup file. It uses a command called view that is available on SolvNet, Doc Id 014947.

The alias vman will not work if the "wish" executable, the main executable in the Tk package, is not installed and made available in your lab environment

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Question 25. How will you modify **check_timing** to add a check to validate that all clocks are propagated?

.....

7. Quit *PrimeTime*.

Task 5. Re-Execute the Run Script to Fix the Problem

1. Add two variables to **./scripts/orca_pt_variables.tcl** to accomplish the following two things.
   - Add to the default checks performed by **check_timing** the check that will flag ideal clocks.
   - All created clocks will be created as propagated clocks.
2. Save the file.
3. Invoke PrimeTime in order to identify any typos of application variable names in this same file.

Question 26. Describe how you will verify there are no typos in this file?

.....

Question 27. Which application variable was mistyped (there is at least one) and what is the correct variable name?

.....

4. Fix all typos in this file and save the file.
5. Quit PrimeTime.
6. Execute the run script **./RUN.tcl** from the **lab9_clocks** Unix directory

   Log the results using Unix command **tee -i**.

   Name the log file run.log.

```
unix> pt_shell -f ./RUN.tcl | tee -i run.log
```

Question 28. How will you know if an error occurs during the execution of the run script?

.....

7. Invoke PrimeTime and restore the newly saved session in the Unix directory **./orca _savesession**.

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8. Use the appropriate commands to confirm the information below:
   - The **out_setup** violations have been reduced.
   - All clocks are propagated.
   - Execute **check_timing** to confirm it is performing its default checks in addition to the check for ideal clocks.
   - The source latency is now calculated for **SD_DDR_CLK.**
   - The timing report to **sd_DQ[3]** includes this calculated source latency.

   There will be additional violations (more setup violations as well as out_hold violations) that you can ignore.

9. Quit PrimeTime.

This completes lab 8. Return to lecture.

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Answers / Solutions

Question 1. How many clocks are in this design and how many of these are generated?

This information can be gathered from **report_clock**, or using the following commands.

```
pt_shell> sizeof_collection [all_clocks]
6
pt_shell> sizeof_collection [get_generated_clocks *]
3
```

Question 2. Which input ports have defined, master clocks?

```
pt_shell> rpt_clock_ports
Port Name        Direction    Clock Name      Is Generated
-----------------------------------------------------------
  pclk            in           PCI_CLK         false
  sys_clk         in           SYS_CLK         false
  sdr_clk         in           SDRAM_CLK       false
  sd_CK           out          SD_DDR_CLK      true
  sd_CKn          out          SD_DDR_CLKn     true
```

Question 3. Which output ports have defined, outgoing clocks?

From the same report, **sd_CK** and **sd_CKn**.

Question 4. Are the clocks propagated or ideal?

Use **report_clock** to see that all the design clocks are ideal.

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Question 5. Which 3 clock pairs have constrained timing paths?

```
pt_shell> check_timing -over clock_crossing -verbose
Information: Checking 'clock_crossing'.
Information: There are 4 clocks having domains interacting.

*                        all paths are false paths
#                        part of paths are false paths

From Clock               Crossing Clocks
-------------------------------------------------------------
PCI_CLK                  SYS_CLK*
SDRAM_CLK                SD_DDR_CLK#, SYS_CLK*
SYS_2x_CLK               SDRAM_CLK*, SYS_CLK
SYS_CLK                  PCI_CLK*, SDRAM_CLK*, SYS_2x_CLK
```

Question 6. What is the master clock for **SYS_2x_CLK**?

SYS_CLK

Question 7. SYS_2x_CLK is defined on which pin/port (its "source")?

I_CLOCK_GEN/I_CLKMUL/CLK_2X (You may have to drag the window containing the matrix out of the way in order to see the source pins)

Question 8. The master clock for SYS_2x_CLK is defined on which pin/port?

sys_clk

Question 9. What port is connected to the select pin of the MUX I_CLOCK_GEN/U20?

test_mode

Question 10. Does seeing the schematic give you insight into the clocking scheme for test?

Yes – two clocks come into the mux: one from the clock generator, one directly from the port. The test_mode port controls the select line, making it possible to bypass the clock generator during test mode and letting the design be driven directly from the port.

Question 11. What clocks drive I_CLOCK_GEN/U10?

SYS_CLK and SYS_2x_CLK

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Question 12. What port drives the select line to I_CLOCK_GEN/U10?

power_save

Question 13. From the abstract clock graph window, is it possible to display the same clock schematic you displayed in the clock analyzer?

Yes. Select the clock SYS_2x_CLK (you may have to zoom in to select just the clock), then press right mouse button and select `Schematic for Selected Clocks`. You may have to expand input or output stubs (by double clicking on them) to get the exact same schematic.

Question 14. What percent of the capture delay comes from CRP?
4.20%

Question 15. Is this percent representative of all designs?
No, this number is dependent on the particular design and on the particular path.

Question 16. How wide is the arrival window for the buffer output pin?
It is 2.86939 minus 2.39925, or about .47.

Question 17. Does this match what we saw earlier in the data arrival data required section of the path inspector?
Yes.

Question 18. What can you add to the waveforms by clicking the right mouse button in the waveform window?
You can add input pins and output pins, allowing you to see the detail for the whole path.

Question 19. How many, and what kind of violations does **ORCA** have?

There are 23 *setup* violations, 53 *hold* violations, and 32 *out_setup* violations.

Question 20. How will you identify the endpoint port which has the worst slack for out_setup?

```
pt_shell> report_timing -path short -to [all_outputs]
  OR
pt_shell> page_on
pt_shell> report_analysis_coverage -status violated -check out_setup
Type of Check        Total          Met        Violated        Untested
```


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```
-----------------------------------------------------------------------
out_setup          75        43 ( 57%)        32 ( 43%)        0 (  0%)
-----------------------------------------------------------------------
All Checks         75        43 ( 57%)        32 ( 43%)        0 (  0%)

Constrained                    Related    Check
Pin                            Pin        Type
Slack
-----------------------------------------------------------------------
sd_DQ[6]                       SD_DDR_CLK   out_setup        -1.4006
sd_DQ[5]                       SD_DDR_CLK   out_setup        -1.4006
sd_DQ[4]                       SD_DDR_CLK   out_setup        -1.4006
```

Question 21. Which clocks (launch and capture) are involved in this violation?

```
pt_shell> report_timing -to sd_DQ[6] -path short
****************************************
  Startpoint: sdr_clk (clock source 'SDRAM_CLK')
  Endpoint: sd_DQ[6] (output port clocked by SD_DDR_CLK)
  Path Group: SD_DDR_CLK
  Path Type: max
  Min Clock Paths Derating Factor : 0.9000

  Point                                    Incr       Path
  ---------------------------------------------------------------
  clock SDRAM_CLK (fall edge)            3.7500     3.7500
  clock source latency                   0.0000     3.7500
  sdr_clk (in)                           0.0000     3.7500 f
  ...
  sd_DQ[6] (inout)                       4.4006     8.1506 f
  data arrival time                                 8.1506

  clock SD_DDR_CLK (rise edge)           7.5000     7.5000
  clock reconvergence pessimism          0.0000     7.5000
  output external delay                 -0.7500     6.7500
  data required time                                6.7500
  ---------------------------------------------------------------
  data required time                                6.7500
  data arrival time                                -8.1506
  ---------------------------------------------------------------
  slack (VIOLATED)                                 -1.4006
```

Question 22. Why has PrimeTime not calculated source latency for the outgoing clock **SD_DDR_CLK**?

The clocks (specifically the master clock) must be propagated for *PrimeTime* to calculate the source latency for generated clocks. All clocks in this design are ideal.

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Question 23. What is the name of this variable?

```
pt_shell> aa propagate
********* Commands **********
 remove_propagated_clock # Remove a propagated clock specification
 set_propagated_clock # Specify propagated clock latency
********* Variables **********
case_analysis_propagate_through_icg = "false"
timing_all_clocks_propagated = "false"
timing_clock_gating_propagate_enable = "false"
timing_propagate_interclock_uncertainty = "false"
timing_propagate_through_non_latch_d_pin_arcs = "false"
```

Question 24. Using a man page, explain what this variable will do?

All clocks created after this variable is set to true will be created as propagated clocks.

The clocks will be set to propagated in the next task.

Question 25. How will you modify **check_timing** to add a check to validate that all clocks are propagated?

The added check is named **ideal_clocks**. You can add this check to the variable `timing_check_defaults` using `lappend` such that it is executed automatically with `check_timing`.

This will be done in the next task.

```
# Answers for TASK 5 STEP 1
# Add the following to ./scripts/orca_pt_variables.tcl
lappend timing_check_defaults ideal_clocks
set timing_all_clocks_propagated true
```

Question 26. Describe how you will verify there are no typos?

Source this file in PrimeTime. If a new variable message occurs (`CMD-041`), there is a typo!

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Question 27. Which application variable was mistyped (there is at least one) and what is the correct variable name?

The variable **link_create_blackboxes** should be **link_create_black_boxes**.

Question 28. How will you know if an error occurs during the execution of the run script?

Errors will terminate the execution of the run script due to the following settings:

```
sh_continue_on_error=false
sh_script_stop_severity=E
```

If the script completes, there were no errors.

```
# Answers for Task 5 Step 8
# Out_setup violations should be reduced to 9 violations.
pt_shell> report_analysis_coverage

# All clocks should be propagated
pt_shell> report_clock

# The command check_timing does not flag ideal clocks
pt_shell> check_timing
Information: Checking 'no_clock'.
Information: Checking 'no_input_delay'.
Information: Checking 'partial_input_delay'.
Information: Checking 'ideal_clocks'.
. . .
# The source latency is being calculated for SD_DDR_CLK
pt_shell> report_clock -skew SD_DDR_CLK

# The source latency is applied to the timing report to SD_DQ[6]
pt_shell> report_timing -to sd_DQ[6] -path short
  Startpoint: sdr_clk (clock source 'SDRAM_CLK')
  Endpoint: sd_DQ[6] (output port clocked by SD_DDR_CLK)
  Last common pin: sdr_clk
  Path Group: SD_DDR_CLK
  Path Type: max
  Min Clock Paths Derating Factor : 0.9000

  Point                                    Incr       Path
  ---------------------------------------------------------------
```


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```
clock SDRAM_CLK (fall edge)          3.7500     3.7500
clock source latency                 0.0000     3.7500
sdr_clk (in)                         0.0000     3.7500 f
...
sd_DQ[6] (inout)                     4.4006     8.1506 f
data arrival time                               8.1506

clock SD_DDR_CLK (rise edge)         7.5000     7.5000
clock network delay (propagated)     2.9149    10.4149
clock reconvergence pessimism        0.0000    10.4149
output external delay               -0.7500     9.6649
data required time                              9.6649
-------------------------------------------------------------
data required time                              9.6649
data arrival time                              -8.1506
-------------------------------------------------------------
slack (MET)                                     1.5143
```


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10 Timing Analysis Flow

Learning Objectives

After completing this lab, you should be able to:

- Check an existing PrimeTime session
  - Search path components
  - Qualities of input files (netlists, libraries, back annotation, constraints)
  - Summary of informational, warning, and error messages
- Debug scripts
  - Display executing commands and results
  - Display additional debugging information
  - Redirect output of session and of specific commands to a file
- Improve PrimeTime runs
  - Performance
  - Convenience
- Save a session to be shared with others

Lab Duration: 60 minutes

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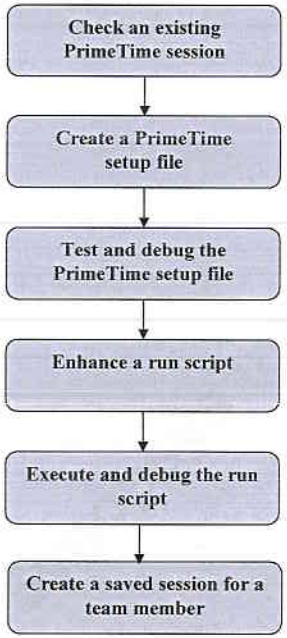
Introduction



Relevant Files and Directories

All files for this lab are located in the **`lab10_flow`** directory under your home directory.

`lab10_flow/`	Current working directory
`common_setup.tcl`	multi-tool shared setup file
`pt_setup.tcl`	tool-specific PrimeTime setup file
`pt_scripts/`	Run file directory
`pt.tcl`	Run file
`.synopsys_pt.setup`	automatically-read PT setup file.
`orca_savesession`	Saved session directory

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Instructions

Task 1. Check an Existing PrimeTime Session

In this task, you will validate the inputs that have been read into PrimeTime: the current design and libraries, the backannotation and constraints.

1. Change into the working directory for this lab:

```
unix% cd lab10_flow
```

You will stay in this directory for the entire lab.

2. Bring up PrimeTime and restore a saved session called `orca_savesession`. Please ignore the `CMD-029` warning – you will handle it later.

```
unix% pt_shell
pt_shell> restore_session orca_savesession/
```

3. Verify that the current design is your top-level module: ORCA

```
pt_shell> current_design
```

4. Compare the unix paths of the libraries to what has been read into PrimeTime

```
pt_shell> printvar search_path
pt_shell> printvar link_path
pt_shell> list_libraries
```

Question 1. Have the 4 libraries in the link_path been successfully read into PrimeTime?

.....

Question 2. Which library defines the defaults for time units, operating conditions, and other delay calculation information?

.....

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Question 3. What time unit is used?

.....

5. Verify that the nets are completely annotated.

```
pt_shell> report_annotated_parasitics
```

Question 4. Are there any nets that are not annotated?

.....

Question 5. What option to `report_annotated_parasitics` would be good to use as a 'next step' in debugging the missing nets?

.....

6. Verify that the design is completely constrained.

```
pt_shell> check_timing
```

Question 6. What option to `check_timing` would be good to use as a 'next step' in debugging the missing constraints?

.....

7. Verify that the checks in your cells are completely exercised; look at possible causes for your findings.

```
pt_shell> report_analysis_coverage
pt_shell> report_case_analysis
```

Question 7. Is it logical that many of your timing checks are untested?

.....

Question 8. What options to `report_analysis_coverage` might be useful to find out why the setup checks are untested?

.....

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Task 2. Create a Setup File

1. Open a text editor in the **lab10_flow** Unix directory to modify an automatically-read setup file for PrimeTime.

   Question 9. What is the name of the automatically-read setup file?

   .....

2. Create an alias for `report_timing` with your favorite options. (Common `report_timing` favorites are `-capacitance -transition_time -nets -input_pins -path full_clock_expanded -significant_digits 4`).
3. Suppress the warning message `CMD-029`, which gives you unneeded information about an alias.
4. Add comments describing the purpose of your aliases and suppressed messages.

   Question 10. In Tcl, how are comments designated?

   .....

5. Have a message echoed to the screen when pt_shell completes executing the setup file.

```
echo "setup file completed successfully, dear student"
```

6. Save your setup file.

Task 3. Test the Setup File

1. Execute the setup file in three ways:
   a. Invoke PrimeTime from the **lab10_flow** Unix directory.
   b. Observe the execution of the setup file by executing the setup file from within PrimeTime, echoing commands to the screen.
   c. Capture the effects of executing the setup file by redirecting output to a file.

   Question 11. What commands did you use?

   .....

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.....

Question 12. What is one advantage of using the redirect command?

.....

Question 13. Is it possible to redirect output to both a file and the screen at the same time?

.....

2. Use the alias you created to do a timing report. That is, do a `restore_session orca_savesession` and execute your alias.

   Question 14. Did it work as you hoped?

   .....

3. Use one of the Tcl procedures executed in the setup file.

   Search Synopsys commands and variables for those containing the word **message**.

```
pt_shell> aa message
```

   Question 15. Which command will display the currently suppressed message ID's?

   .....

4. Verify that the message **CMD-029** is being suppressed.

   Question 16. What additional messages are also being suppressed?

   .....

5. Quit PrimeTime.

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Task 4. Enhance a run script

In this task, you will add commands and variables to the run script to help you in debugging and in optimizing performance, and you will save a session for future interactive exploration.

1. With a text editor, open a run script called `./pt_scripts/pt.tcl`
2. Capture the location of all timing updates in your run file by setting `timing_update_status_level` to `high`
3. Capture runtime information by placing `start_profile` and `stop_profile` commands at beginning and end of your run script.
4. Add as the last step in the run script the command to exit from PrimeTime.
5. Capture a summary of all errors, warnings, and informational messages by placing a `print_message_info` command just before the last step of your script.
6. Just before the `print_message_info` command, save the session to a directory called `my_savesession`.
7. Add comments throughout the run script.
8. Save the file.

Task 5. Execute and Debug the Run Script

1. Execute the run script and log the results using the Unix command **`tee`**. Name the log file **`run.log`**.

```
UNIX> pt_shell -f ./pt_scripts/pt.tcl | tee -i run.log
```

Question 17. How will you know if an error occurs during the execution of the run script?

.....

2. If there are any errors, address these first before moving on to the next step.
3. Evaluate your log file. With a text editor, open your log file. Search for the update timing output (`UITE-214`), for `print_message_info` output, and for `quit` output. Then, in your `profile` directory, browse `tcl_profile_self_only_sorted_by_cpu_time`.

Question 18. What step required the most CPU time?

.....

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Question 19. What commands were responsible for `UITE-214` messages?

.....

.....

Question 20. Can any of the timing updates be avoided?

.....

Question 21. Why might the `quit` command output be a good place to start before reviewing your log file?

.....

4. Quit PrimeTime.

Task 6. Create a Saved Session for a Team Member

1. In the empty Unix directory **joe_project_dir**, copy (and modify) the appropriate files such that another team member on a completely independent system will be able to use this directory to perform STA on **ORCA**. Assume that the Joe already has the needed libraries and that they are in a `ref` directory, just as they are on your system.

Question 22. What Unix directory did you copy to **./joe_project_dir**?

.....

Question 23. What is the one file you modified so that PrimeTime can find the correct library files?

.....

Question 24. What is one reason why copying the PrimeTime .synopsys_pt.setup file to **./joe_project_dir** is helpful?

.....

2. To test what you have just created, invoke PrimeTime from the directory **joe_project_dir.**

Restore the saved session for further debugging of **ORCA**.

Question 25. What is one way to find the command to restore a PrimeTime session?

.....

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Question 26. From the informational messages generated from the `restore_session` command, where were the library files found that were loaded into PrimeTime?

.....

3. Quit PrimeTime.

Task 7. Optional: Search SNUG for a Relevant Paper

1. Open a web browser and follow the link to the "Synopsys Users Group" from the www.synopsys.com web page.
Alternatively, there is a link to "SNUG – Synopsys Users Group" on the left hand side of the SolvNet home page.

```
unix% firefox &
```

2. Search for prize-winning presentations from the 2014 Silicon Valley conference. (click on Silicon Valley, then on Download Proceedings, then on Sort proceedings by Awards). Glance at the conclusion to the 2nd Place for Best Paper: Advanced Node Random Device Variability and Margining in Characterization and STA.

Question 27. What technologies are compared?

.....

This completes Lab 10. Return to lecture.

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Answers / Solutions

Question 1. Have the 4 libraries in the link_path been successfully read into PrimeTime?

Yes – the four libraries specified with `link_library` are the same as those displayed with `list_libraries`.

Question 2. Which library defines the defaults for time units, operating conditions, and other delay calculation information?

cb13fs120_tsmc_max – the asterisk to the left of the library (displayed with `list_libraries`) indicates 'main library'

Question 3. What time unit is used?

```
pt_shell> report_lib  cb13fs120_tsmc_max
****************************************
Report : library
Library: cb13fs120_tsmc_max
Version: K-2015.06-SP2
Date   : Wed Oct 28 09:37:46 2015
****************************************

Time Unit                    : 1 ns
Capacitance Unit             : 1 pF
```

```
pt_shell> report_units
****************************************
Report : units
Design : ORCA
Version: K-2015.06-SP2
Date   : Wed Oct 28 09:33:27 2015
****************************************

Units
--------------------------------------------
-
Capacitive_load_unit         : 1e-12 Farad
Current_unit                 : 1e-06 Amp
Resistance_unit              : 1000 Ohm
Time_unit                    : 1e-09 Second
```


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Question 4. Are there any nets that are not annotated?

Yes: there are some internal driverless nets and some boundary pin-to-pin nets that are not annotated.

Question 5. What option to `report_annotated_parasitics` would be good to use as a 'next step' in debugging the missing nets?

`report_annotated_parasitics -help` shows options; the `-list_not_annotated` option shows the net names; you might want to focus in on the pin-to-pin nets by using the `-pin_to_pin_nets` option.

Question 6. What option to `check_timing` would be good to use as a 'next step' in debugging the missing constraints?

`check_timing -help` displays a list of options: the `-verbose` option lists the unconstrained endpoints

Question 7. Is it logical that many of your checks are untested?

Yes – test_mode, scan_en, and power_save are constrained off – performing STA on additional modes might enable more checks to be tested.

Question 8. What options to `report_analysis_coverage` might be useful to find out why the setup checks are untested?

```
report_analysis_coverage \
-status_details untested \
-check_type setup
```

Question 9. What is the name of the setup file?

The setup file must be named **.synopsys_pt.setup** such that PrimeTime will find and execute this file automatically every time the tool is invoked.

Question 10. In Tcl, how are comments designated?

Use a # character at the beginning of each line that is a comment

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Question 11. What commands did you use?

`pt_shell` automatically runs `.synopsys_pt.setup` when it is invoked
`source -echo -verbose .synopsys_pt.setup`
`redirect myfile { source -echo -verbose .synopsys_pt.setup}`

Question 12. What is one advantage of using the redirect command?

It is good for batch jobs, and the file is easy to browse using an editor.

Question 13. Is it possible to redirect output to both a file and the screen at the same time?

Yes. `redirect -help` shows options; the `-tee` option does what you want.

Question 14. Did it work as you hoped?
If not, revise your setup file and test again.

Question 15. Which command will display the currently suppressed message ID's?

The command **print_suppressed_messages.**

Question 16. What other message is also being suppressed?

Messages may be suppressed by the setup file in your admin directory or built into the PrimeTime tool itself. Note that if you unsuppressed **CMD-029** in your setup file, it will not be listed below.

```
pt_shell> print_suppressed_messages
The following 4 messages are suppressed:
        CMD-029, RC-004, RC-009, SVR-2
```

You should have expected that the message **CMD-029** occurred and was suppressed at least once. Any other warnings or errors that have occurred during your session should also be listed.

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Question 17. How will you know if an error occurs during the execution of the run script?

Any errors will terminate the script in the middle of execution. If the script completes, no errors occurred during execution of the run script.

Question 18. What step required the most CPU time?
For this lab, sourcing the constraints took the most CPU time.

Question 19. What commands were responsible for `UITE-214` messages?
update_timing –full and a call to a macro PLL_SHIFT which causes timing to be recalculated

Question 20. Can any of the timing updates be avoided?
Yes, there is an `update_timing` both before and after `set_propagated_clocks` – the one before `set_propagated_clocks` is unnecessary. One `update_timing` command is in pt.tcl – the other is in a constraint file (orca_pt_other.tcl) sourced by orca_pt_constraints which is sourced by pt.tcl

Question 21. Why might the `quit` command output be a good place to start before reviewing your log file?
It gives a high-level summary of potential trouble spots: messages, both warning and information, timing updates, and performance statistics.

Question 22. What is the Unix directory you copied to **./joe_project_dir**?

The Unix directory and its contents: **./my_savesession**.

Question 23. What is the one file you modified so that PrimeTime can find the correct library files?

Modify the file **joe_project_dir/my_savesession/lib_map** such that the library file Unix path names point to the library files in the ref directory. Because you do not know the absolute path names on Joe's machine, change the absolute path names to relative path names: `../../ref/libs/your_lib.db`.

Modifying the lib_map file is not necessary if all users are sharing a common library directory in your work environment.

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Question 24. What is one reason why copying the PrimeTime setup file to **./joe_project_dir** is helpful?

The setup file contains useful aliases and executes several useful Tcl procedures. It also suppresses messages that have been researched and deemed unimportant. If you copy the setup file – do not forget to copy the Unix directory that contains the Tcl procedures!

Question 25. What is one way to find the command to restore a PrimeTime session?

Here are four:

- Invoke PrimeTime and use the Tcl procedure **aa restore**.
- Use **help *restore***.
- Use the provided job aids
- Use command-line expansion; that is, type `rest`, and then press the tab key.

Question 26. From the informational messages generated from the `restore_session` command, where were the library files found that were loaded into PrimeTime?

```
pt_shell> restore_session orca_savesession
.....
Loading db file '../../ref/libs/sc_max.db'
Loading db file '../../ref/libs/io_max.db'
Loading db file '../../ref/libs/mem_max.db'
Loading db file '../../ref/libs/special.db'
.....
```

Question 27. What technologies are compared?

SPICE, OCV, AOCV, and POCV.

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11 Specific Cases

Learning Objectives

After completing this lab, you should be able to:

- Apply user specified annotated delays to explore time borrowing with latches
- Debug **PTE-070** messages regarding non-unate cells on the clock path

Lab Duration:
30 minutes

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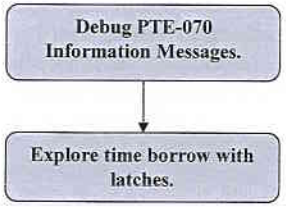
Overview



Relevant Files and Directories

All files for this lab are located in the **lab11_specific** directory under your home directory.

lab11_specific/	Current working directory
orca_savesession/	Saved session for ORCA
logs/	Log files from run script

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Instructions

Task 1. Debug PTE-070 Information Messages

1. Invoke PrimeTime from the **lab11_specific** Unix directory.

   Restore the session saved under **./orca_savesession**.

2. Shown below is the full message regarding a non-unate path on the clock network.

   In the next step, you will be asked to generate a timing report through this pin. In order to copy and paste and avoid typos – either find this message in the log file from another terminal window or use the Unix command **grep** from within PrimeTime as shown below.

```
# From ./logs/run.log
Information: A non-unate path in clock network detected.
Propagating both inverting and noninverting senses of clock
'SDRAM_CLK' from pin
'I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_0/Z'. (PTE-070)
```

```
pt_shell> sh grep -A 1 -B 1 PTE-070 logs/run.log
```

Note: The command **sh** (or alternatively **exec**) allows you to execute Unix commands from within the PrimeTime shell.

3. Generate a timing report for setup through the above pin and answer the following questions.

   The following alias has been created in the PrimeTime setup file and will generate a timing report in a pop-up window with a scroll bar using the view utility found on SolvNet, Doc Id 014947.

```
pt_shell> vrt -through <through pin>
```

Question 1. Which lines in the timing report did you use to validate it is for setup and the timing path start point is the source for the clock **SDRAM_CLK**?

.....

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Question 2. How does this timing report confirm that the pin in the warning above is on a data path (i.e. a clock source being used and constrained as a data path) and not on a clock path?

.....

Question 3. Which sense is propagated through the above pin (i.e. positive unate or negative unate)? Look for a small arrow in the timing report which will locate the specific pin of interest.

.....

4. Generate at least one additional timing report to show the use of a negative unate timing arc through the pin of interest.

   Question 4. Which lines in the timing report did you use to validate it is for setup, the timing path start point is the source for the clock **SDRAM_CLK** and that the timing arc is negative unate for the pin of interest?

   .....

   Question 5. Explain why this warning can be ignored (and suppressed) for these timing paths?

   .....

5. Do not quit PrimeTime.

Task 2. Explore Time Borrow and Latches

There is only one latch in this design.

1. Use the following commands to find it:

   Take advantage of command and option completion with the tab key.

```
pt_shell> all_registers -level_sensitive
pt_shell> !! -clock_pin
pt_shell> all_registers -level_sensitive -data_pins
```

Question 6. What is the name of the clock pin for this latch?

.....

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Question 7. What are the names of the three data pins?

.....

2. Generate a timing report starting at the latch for setup time (be specific by using the clock pin as the start point and not just the cell name!).

   This lab will refer to this timing report as "path segment #2".

   The function of this latch in the **ORCA** design is to generate a clock gating signal to turn on and off the clock **SYS_CLK**.

   Question 8. Describe how you know this latch is not experiencing time borrow from the previous stage?

   .....

3. Generate a timing report for the previous stage (this lab will refer to this timing report as "path segment #1").

   Use the **D** input pin of the latch as the end point of this timing path.

   Question 9. How much more time can path segment #1 take before it would start borrowing time from path segment #2?

   .....

4. Force path segment #1 to borrow time from path segment #2 by annotating a net delay of 4ns as shown below:

```
# Use cut and paste to avoid typos on the pin name
pt_shell> set_annotated_delay -net 4 \
          -to I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/D
```

5. Generate the timing report for path segment #1 again (take advantage of the up and down arrows to scroll through the history event list).

   Question 10. How much time is path segment #1 borrowing from path segment #2?

   .....

   Question 11. What is the slack for path segment #1?

   .....

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6. Re-generate the timing report for path segment #2. Before you can do that, you have to perform a full timing update!

```
pt_shell> update_timing -full
```

   Note: The start point of the timing path will now be the **D** pin of the latch (not the clock pin as used before) because you are interested in reporting the timing path that includes time borrow.

```
pt_shell> report_timing -from \
          I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/D
```

   Question 12. Does the time given to path segment #1 now match your expectations?

   .....

7. Change the latch behavior for transparency; that is, make it transparent when data arrives between the opening and closing edges of the clock.

```
set_app_var timing_enable_through_paths true
```

8. Repeat your timing report to the latch D pin. Notice that, even though the latch is transparent, you can still specify the D pin as an endpoint.

```
report_timing -to \
"I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/D"
```

   Question 13. What is the startpoint?

   .....

9. Do a timing report FROM the startpoint you just identified.

```
report_timing -from \
I_ORCA_TOP/I_PARSER/blender_clk_en_reg/CP
```

   Question 14. What is the path endpoint?.

   .....

   Question 15. What are the transparency open and close edges?.

   .....

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Question 16. Did data arrive between them?.

.....

Question 17. Was there time borrowing?.

.....

Question 18. Was slack positive?.

.....

Question 19. Quit PrimeTime.

This completes Lab 11. Return to lecture.

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Answers / Solutions

Question 1. Which lines in the timing report did you use to validate it is for setup and the timing path start point is the source for the clock **SDRAM_CLK**?

```
pt_shell> report_timing -through I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_0/Z
  Startpoint: sdr_clk (clock source 'SDRAM_CLK')
  Endpoint: sd_DQ[0] (output port clocked by SD_DDR_CLK)
  Path Group: SD_DDR_CLK
  Path Type: max
```

Question 2. How does this timing report confirm that the pin in the warning above is on a data path (i.e. a clock source being used and constrained as a data path) and not on a clock path?

If no report was generated ("path is unconstrained"), this pin is on a clock path. Because a timing report was generated, this pin is on a data path.

Question 3. Which sense is propagated through the above pin (i.e. positive unate or negative unate)? Look for a small arrow in the timing report which will locate the specific pin of interest.

A positive unate timing arc (fall to fall) is reported through this pin.

```
# Shown are the relevant lines in the data path

I_ORCA_TOP/I_SDRAM_IF/buffd7G5B2I36/Z (buffd7)          0.1634 &   4.8748 f
I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_0/Z (mx02d4) <-     0.5431 &   5.4179 f
```

Question 4. Which lines in the timing report did you use to validate it is for setup, the timing path start point is the source for the clock **SDRAM_CLK** and that the timing arc is negative unate for the pin of interest?

```
pt_shell> pt_shell> report_timing \
            -rise_through    I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_0/Z
Startpoint: sdr_clk (clock source 'SDRAM_CLK')
Endpoint: sd_DQ[0] (output port clocked by SD_DDR_CLK)
Path Group: SD_DDR_CLK
Path Type: max
```


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...

```
I_ORCA_TOP/I_SDRAM_IF/buffd7G5B2I36/Z (buffd7)          0.1634 &   4.8748 f
I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_0/Z (mx02d4) <-     0.4928 &   5.3676 r
```

Question 5. Explain why this message can be ignored for these timing paths?

The message indicates that both senses of the clock will be used when propagating the clock through this mux – this is the default behavior. However, because the clock is being used as data, PrimeTime actually propagates both senses (both positive and negative unate), even in older versions of PrimeTime. This is what is desired and therefore this information message can be ignored.

Question 6. What is the name of the clock pin for this latch?

```
pt_shell> all_registers -level_sensitive -clock_pins
{"I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/EN"}
```

Question 7. What are the names of the three data pins?

```
pt_shell> all_registers -level_sensitive -data_pins
{"I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/D",
"I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/SC",
"I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/SD"}

# For step 2, generate a timing report for path segment 2
pt_shell> report_timing -from
I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/EN
```


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Question 8. Describe how you know this latch is not experiencing time borrow from the previous stage?

If this latch were experiencing time borrow, there would be a line in the report stating the amount of time given to the start point (i.e. to the previous stage). This line is not present in this timing report.

```
# For step 3, generate a timing report for path segment 1
pt_shell> report_timing -to
I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/D
```

Question 9. How much more time can path segment #1 take before it would have to start borrowing time from path segment #2?

It can take 3.465ns more before it would start borrowing time from path segment #2 (equivalent to the positive slack).

Question 10. How much time is path segment #1 borrowing from path segment #2?

Path segment #1 is borrowing 0.5306ns from path segment #2. This is noted in the data required time section of the timing report.

Question 11. What is the slack for path segment #1?

The slack is zero. PrimeTime borrows exactly as much as is needed to make the slack equal zero.

Question 12. Does the time given to path segment #1 match your expectations?

Yes – the time given to start point in the timing report for path segment #2 will match the time borrowed in the timing report for path segment #1.

```
pt_shell> report_timing -from I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/D
pt_shell> report_timing -to I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/D
```


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Question 13. What is the startpoint?

I_ORCA_TOP/I_PARSER/blender_clk_en_reg/CP

Question 14. What is the path endpoint?

I_ORCA_TOP/I_BLENDER/U794/A (a gating check)

Question 15. What are the transparency open and close edges?

6.6454 and 10.4433

Question 16. Did data arrive between them?

yes – at time 7.1759

Question 17. Was there time borrowing?

no

Question 18. Was slack positive?.

yes

The timing report through the transparent latch is on the next page:

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```
pt_shell> report_timing -from I_ORCA_TOP/I_PARSER/blender_clk_en_reg/CP
Startpoint: I_ORCA_TOP/I_PARSER/blender_clk_en_reg
              (rising edge-triggered flip-flop clocked by SYS_CLK)
  Endpoint: I_ORCA_TOP/I_BLENDER/U794
              (rising clock gating-check end-point clocked by SYS_CLK)
  Last common pin: I_CLK_SOURCE_SYS_CLK/Z
  Path Group: **clock_gating_default**
  Path Type: max
  Min Clock Paths Derating Factor : 0.9000

  Point                                                    Incr       Path
  ---------------------------------------------------------------------------
  clock SYS_CLK (rise edge)                              0.0000     0.0000
  clock network delay (propagated)                       2.7139     2.7139
  I_ORCA_TOP/I_PARSER/blender_clk_en_reg/CP (sdcrb1)     0.0000     2.7139 r
  I_ORCA_TOP/I_PARSER/blender_clk_en_reg/Q (sdcrb1)      0.4621 &   3.1759 r
  I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/D (slnlq1)     4.0000 *   7.1759 r
  ---------------------------------------------------------------------------
   transparency window #1

  clock SYS_CLK (fall edge)                                         4.0000
  clock latency                                          2.3250     6.3250
  clock reconvergence pessimism                          0.3203     6.6454
  transparency open edge                                            6.6454

  clock SYS_CLK (rise edge)                                         8.0000
  clock latency                                          2.3596    10.3596
  clock reconvergence pessimism                          0.3203    10.6800
  library setup time                                    -0.2367    10.4433
  transparency close edge                                          10.4433

  available borrow at through pin                        3.2674
  ---------------------------------------------------------------------------
  I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/D (slnlq1)     0.0000     7.1759 r
  I_ORCA_TOP/I_BLENDER/latched_clk_en_reg/Q (slnlq1)     0.3164 H   7.4924 r
  I_ORCA_TOP/I_BLENDER/U794/B1 (ora21d4)                 0.0263 &   7.5186 r
  data arrival time                                                 7.5186

  clock SYS_CLK (rise edge)                              8.0000     8.0000
  clock network delay (propagated)                       1.6425     9.6425
  clock reconvergence pessimism                          0.2160     9.8585
  I_ORCA_TOP/I_BLENDER/U794/A (ora21d4)                             9.8585 r
  clock gating setup time                               -0.2000     9.6585
  data required time                                                9.6585
  ---------------------------------------------------------------------------
  data required time                                                9.6585
  data arrival time                                                -7.5186
  ---------------------------------------------------------------------------
  slack (MET)                                                       2.1398
```


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12 Path-Based Analysis

Learning Objectives

After completing this lab, you should be able to:

- Improve PrimeTime accuracy using
  - Interactive PBA on a collection
  - Exhaustive PBA

Lab Duration:
20 minutes

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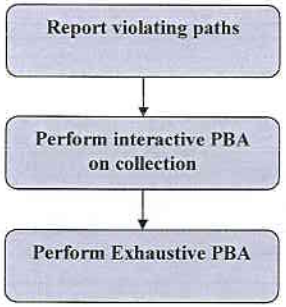
Introduction



Relevant Files and Directories

All files for this lab are located in the **`lab12_pba`** directory under your home directory.

`lab12_pba/`	Current working directory
`.synopsys_pt.setup`	automatically-read PT setup file
`orca_savesession`	Saved session directory

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Instructions

Task 1. Report violating paths in the current design

In this task, we will report the worst violating slacks calculated during worst_slew propagation for the design.

1. Change into the working directory for this lab:

```
unix% cd lab12_pba
```

2. Bring up PrimeTime and restore a saved session called `orca_savesession`.

```
unix% pt_shell
pt_shell> restore_session orca_savesession/
```

3. Identify the number of setup violations and identify the worst violating path to an endpoint.

```
pt_shell> report_analysis_coverage -check setup \
 -status violated
```

Question 1. How many setup checks are violated?

.....

Question 2. Identify the size of the worst violating slack and the endpoint to which it is reported?

.....

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Task 2. Perform interactive PBA of a collection

In this task, we will interactively recalculate the slack for the worst violating path reported.

1. Determine the worst setup slack violation after the interactive PBA run

```
pt_shell> report_timing -to
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/D -pba_mode path
OR
pt_shell> set path [get_timing_path -to
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/D -pba_mode path]
pt_shell> report_timing $path
```

Question 3. How does report_timing indicate that the path is recalculated?

.....

Question 4. How much is the worst slack violation after interactive path-based analysis –pba_mode path ?

.....

Question 5. Did the slack improve or became worst than the non-PBA slack or there is no change in slack number? Are the results as expected?

.....

Note: PBA does not change the PrimeTime database, so you cannot use `report_analysis_coverage`. In this lab, you can use either `report_timing` or `get_timing_paths` commands with the `-pba_mode` option.

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Task 3. Performing Exhaustive PBA

In this task, we will run the design under exhaustive PBA mode.

1. Identify the top ten setup violations and the violating endpoints for the clock group "SYS_CLK" design.

```
pt_shell> redirect -tee non-pba.rpt { report_timing  \
-path summary -group SYS_CLK -max_paths 10 -nosplit}
```

2. Next, run the exhaustive PBA on the setup violation reported for the design and generate a summary report :

```
pt_shell> redirect -tee pba.rpt { report_timing  \
-path summary -group SYS_CLK  \
-max_paths 10 -nosplit  -pba_mode exhaustive }
```

3. Compare non-pba.rpt and pba.rpt and check :

Question 6. How many violating paths are returned before PBA and after PBA?

.....

```
pt_shell> sizeof_collection [get_timing_paths \
   -slack_less 0 -group SYS_CLK -max_paths 10]
pt_shell> sizeof_collection [get_timing_paths \
   -slack_less 0 -group SYS_CLK -max_paths 10 \
   -pba_mode exhaustive]
```

Question 7. What is the value of worst violation after exhaustive PBA; what is the associated timing path endpoint name?

.....

Question 8. Are there any UITE-480 warnings?

.....

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Answers / Solutions

Note: ***If you run into insurmountable problems in this lab, look at or copy the setup file from the .solutions Unix directory.***

Question 1. How many setup checks are violated?
23

Question 2. Question 2. Identify the size of the worst violating slack and the endpoint to which it is reported?

Slack to endpoint
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/D: -0.7737

Question 3. How does report_timing indicate that the path is recalculated?

The Path Type is followed by 'recalculated'. See the following example:

```
pt_shell> report_timing -to I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/D -
pba_mode path
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -pba_mode path
****************************************

  Startpoint: I_ORCA_TOP/I_BLENDER/s3_op2_reg[18]
              (rising edge-triggered flip-flop clocked by SYS_CLK)
  Endpoint: I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]
              (rising edge-triggered flip-flop clocked by SYS_CLK)
  Path Group: SYS_CLK
  Path Type: max  (recalculated)
```


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Question 4. How much is the worst slack after interactive path-based analysis –pba_mode path?
-0.7711

Question 5. Did the slack improve or became worst than the non-PBA slack or there is no change in slack number? Are the results as expected?

The slack improved as expected.
(Remember that when a path is recalculated, the slack can only remain the same or improve.)

Question 6. How many violating paths are returned before PBA and after PBA?

Before PBA : 9 violating paths

After PBA: 8 violating paths

Question 7. What is the value of worst violation before and after exhaustive PBA; what is the associated timing path endpoint name?

Before PBA :
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/D
-0.7737

After PBA:
I_ORCA_TOP/I_BLENDER/s4_op2_reg[31]/D
-0.7711

Question 8. Are there any UITE-480 warnings?

No, there are no UITE-480 warnings – your exhaustive recalculation is complete.

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13 Best Practices

Learning Objectives

After completing this lab, you should be able to:

- Debug a hold violation with your knowledge of the SPEF parasitics.

Lab Duration: 30 minutes

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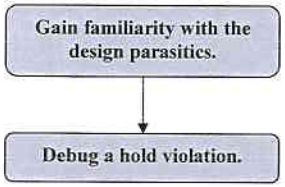
Overview



Relevant Files and Directories

All files for this lab are located in the **`lab13_best_practices`** directory under your home directory.

`Lab13_best_practices/`	Current working directory
`orca_savesession/`	Saved session for ORCA
`logs/`	
`ORCA_EW.log`	Warnings from run script
`run.log`	Log of run
`RUN.tcl`	Run script for this lab

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Instructions

Task 1. Gain Familiarity with the Design Parasitics

You will work from the **lab13_best_practices** Unix directory for this lab. A run script has been executed and the log files are saved under the directory **./logs**. Also, PrimeTime has written the parasitic information to **parasitics_command**.log in the **project** directory.

1. Scroll through **./parasitics_command.log** to answer the following questions.

   Question 1. Describe the category of nets or cells that are missing back annotation?

   .....

2. The next step would be to talk to the people that generated the parasitics and the design constraints.

   Question 2. List a few questions you would ask these people?

   .....

   The following bulleted list contains some answers from your interview. These answers will help when debugging the hold violation in the next task.

- The SPEF file was generated at C_Worst corner (thick metal with high capacitance). Typical SPEF choices are C_Worst and C_Best, which represent worst-case capacitance; RC_Worst and RC_Best, which represent worst-case interconnect RC; and Typical. Additionally, sometimes SPEF is extracted for a Hot or a Cold operating condition, which modifies the resistance.
- Some missing annotations are for the top-level boundary nets. The designer assumed the delays for these nets are zero because these nets connect the ports to the IO pads. There are no such nets on the physical chip (if you do not count the bond wires).
- Other missing annotation is for nets that are remnants after synthesis optimized away their drivers and loads. (Look for the nets in the netlist – you will see the names exist in the module header list, and in the Input declarations, but are not used in any instance. Or, from PrimeTime, execute a **report_net –connections** for any of the driverless nets). These nets have no effect on STA.

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Task 2. Debug a Hold Violation

1. Invoke PrimeTime and restore the session **./orca_savessession**.

   Question 3. How many and what type of timing violations are in this design?

   .....

   Question 4. Name the end point with the worst hold violation to an output port?

   .....

   Question 5. Is CRPR turned on and how would you know?

   .....

2. Generate a timing report to **sd_DQ[0]** for hold.

   The following alias created in the .synopsys_pt.setup file will generate a timing report for hold in a pop-up window with a scroll bar which makes viewing long reports simpler.

```
pt_shell> vrtm -to sd_DQ[0]
# Add input pins to view net and cell delays separately
pt_shell> vrtm -to sd_DQ[0] -input_pins
```

The alias vrtm uses a command called view that is available on SolvNet, Doc Id 014947.

The alias vrtm will not work if the "wish" executable, the main executable in the Tk package, is not installed and made available in your lab environment.

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3. Use the following portion of a man page taken from **report_timing** to help answer the following questions regarding the timing report generated in the last step.

```
pt_shell> vman report_timing
. . . omitted . . .
          Symbol          Annotation
          ------          ----------
          H               Hybrid annotation
          *               SDF back-annotation
          &               RC network back-annotation
          $               RC pi back-annotation
          +               Lumped RC
          <none>          Wire-load model or none
```

Question 6. Which lines in this report can you use to confirm the report is for hold time and is ending at the port **sd_DQ[0]**?

.....

Question 7. Are the boundary net delays along this path set to zero?

.....

4. Set the boundary net delays to zero by executing the following lines:

```
set_annotated_delay 0 -net -to [all_outputs]
set_annotated_delay 0 -net -from [all_inputs]
```

5. Re-execute the STA reports to verify the hold violation is larger and the boundary net delays are now zero.

Question 8. What symbol is now used beside the boundary net delays in **report_timing**?

.....

6. Quit PrimeTime.

This completes Lab13, the last lab for this workshop.

Return to lecture.

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Answers / Solutions

Question 1. Describe the category of nets or cells that are missing annotation.

All the top-level boundary nets are missing annotation. In addition, some internal nets without drivers are missing annotation.

```
# From ./logs/ORCA_EW.log or ./logs/run.log
```

Net Type	Total	Lumped	RC pi	RC network	Not Annotated
Internal nets					
- Pin to pin nets	22566	0	0	22566	0
- Driverless nets	24	0	0	0	24
- Loadless nets	112	0	0	112	0
Boundary/port nets					
- Pin to pin nets	71	0	0	0	71
- Driverless nets	0	0	0	0	0
- Loadless nets	0	0	0	0	0
	22773	0	0	22678	95

Question 2. List a few questions you would ask these people.

What parasitic corner was the SPEF generated for? (Typically, SPEF corners are C_Worst and C_Best, which represent worst-case capacitance; RC_Worst and RC_Best, which represent worst-case interconnect RC; and Typical. Additionally, sometimes SPEF is extracted for a Hot or a Cold operating condition, which modifies the resistance.)

Why are there missing annotations and what assumptions are being made?

With SPEF, you no longer need to ask about constraints used to generate back annotation. That is, with SDF, design constraints affect the extracted SDF, and you must match constraint corners and SDF corners in an STA run. But, with SPEF, design constraints such as clock latency are independent of the extraction process. You no longer have

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to match back annotation and the constraints used to generate it; rather, SPEF is run against all combinations of your design constraints and device process corners.

Question 3. How many and what type of violations are in this design?

```
pt_shell> report_analysis_coverage
Type of Check      Total        Met              Violated          Untested
-----------------------------------------------------------------------------
setup               9629      3575 ( 37%)        13 (  0%)       6041 ( 63%)
hold                9629      3517 ( 37%)        71 (  1%)       6041 ( 63%)
recovery            1316      1210 ( 92%)         0 (  0%)        106 (  8%)
removal             1316      1204 ( 91%)         6 (  0%)        106 (  8%)
min_period            20        20 (100%)         0 (  0%)          0 (  0%)
min_pulse_width     7273      5957 ( 82%)         0 (  0%)       1316 ( 18%)
clock_gating_setup    33        33 (100%)         0 (  0%)          0 (  0%)
clock_gating_hold     33        33 (100%)         0 (  0%)          0 (  0%)
out_setup             75        66 ( 88%)         9 ( 12%)          0 (  0%)
out_hold              75        36 ( 48%)        39 ( 52%)          0 (  0%)
-----------------------------------------------------------------------------
All Checks         29399     15651 ( 53%)       138 (  0%)      13610 ( 46%)
```

Question 4. Name the end point with the worst hold violation to an output port?

```
pt_shell> !! -status violated -check out_hold
Constrained                Related   Check
Pin                        Pin       Type                    Slack
-----------------------------------------------------------------------
sd_DQ[0]                             out_hold              -0.4375
sd_DQ[0]                             out_hold              -0.3956
sd_DQ[10]                            out_hold              -0.3679
```

Note: Two paths to sd_DQ[0] are reported – one for the rising edge of the capture clock, one for the falling edge.

Question 5. Is CRPR turned on and how would you know?

Yes, it is turned on.

```
pt_shell> aa reconvergence
********* Commands **********
Information: No commands matched '*reconvergence*'.
(CMD-040)
********* Variables **********
timing_clock_reconvergence_pessimism = "normal"
timing_remove_clock_reconvergence_pessimism = "true"
```


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Question 6. Which lines in this report can you use to confirm the report is for hold time and is ending at the port **sd_DQ[0]**?

The end point of the timing path is the port **sd_DQ[0]** and the "Path Type" is min.

Question 7. Are the boundary net delays set to zero?

No. The boundary nets are missing parasitics. The designer assumed these delays to be zero. PrimeTime does not make the same assumption. PrimeTime will attempt to calculate the delays for all missing parasitic annotations!

Note in the report below, the delay from the input port has an 'H' symbol and the delay is not zero. The delay to the output port (the final net delay) has no symbol and the delay is not zero. (The H symbol indicates a hybrid annotation; no symbol indicates delay calculation is performed using a WLM).

```
pt_shell> report_timing -delay min -to sd_DQ[0]
. . .
Point                                                    Incr       Path
-----------------------------------------------------------------------------
clock SDRAM_CLK (rise edge)                            0.0000     0.0000
clock source latency                                   0.0000     0.0000
sdr_clk (in)                                           0.0000     0.0000 r
sdr_clk_iopad/CIN (pc3d01)                             0.7275 H   0.7275 r
I_CLOCK_GEN/sdram_clk (CLOCK_GEN)                      0.0000 &   0.7275 r
I_CLOCK_GEN/I_PLL_SD/CLK (PLL)                        -1.1009 H  -0.3734 r
I_CLOCK_GEN/invbdkG1B1I1_2/ZN (invbdk)                 0.1037 &  -0.2696 f
I_CLOCK_GEN/invbdkG1B2I1/ZN (invbdk)                   0.0639 &  -0.2057 r
I_CLOCK_GEN/U22/Z (mx02d2)                             0.2183 &   0.0126 r
I_CLOCK_GEN/invbdkG2B1I1_2/ZN (invbdk)                 0.1012 &   0.1138 f
I_CLOCK_GEN/invbdkG2B2I1/ZN (invbdk)                   0.0655 &   0.1794 r
I_CLOCK_GEN/U18/ZN (invbdk)                            0.0646 &   0.2440 f
I_CLOCK_GEN/U15/ZN (invbdk)                            0.0570 &   0.3010 r
I_CLOCK_GEN/o_sdram_clk (CLOCK_GEN)                    0.0000 &   0.3010 r
I_CLK_SOURCE_SDRAM_CLK/Z (bufbdk)                      0.1654 &   0.4664 r
bufbdaG5B1I6/Z (bufbda)                                0.1626 &   0.6290 r
I_ORCA_TOP/buf_sdram_clk_G5B1I6_1ASTHIRNet744 (ORCA_TOP)
                                                       0.0000 &   0.6290 r
I_ORCA_TOP/I_SDRAM_IF/buf_sdram_clk_G5B1I6_1ASTHIRNet746 (SDRAM_IF)
                                                       0.0000 &   0.6290 r
I_ORCA_TOP/I_SDRAM_IF/buffd7G5B2I36/Z (buffd7)         0.1340 &   0.7629 r
I_ORCA_TOP/I_SDRAM_IF/sd_mux_dq_out_0/Z (mx02d4)       0.3390 &   1.1019 r
I_ORCA_TOP/I_SDRAM_IF/sd_DQ_out[0] (SDRAM_IF)          0.0000 &   1.1019 r
I_ORCA_TOP/sd_DQ_out[0] (ORCA_TOP)                     0.0000 &   1.1019 r
sdram_DQ_iopad_0/PAD (pc3b05)                          2.0287 H   3.1306 r
sd_DQ[0] (inout)                                       0.0430     3.1736 r
data arrival time                                                 3.1736
```


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Use the appropriate switch to separate the net delays from the cell delays and you will see that the boundary nets are not annotated (do not have an & by the delay) and the delays are not zero. They are calculated using wire load models.

```
pt_shell> report_timing -delay min -to sd_DQ[0] -input_pins

. . .
Point                                                  Incr       Path
-----------------------------------------------------------------------
clock SDRAM_CLK (rise edge)                          0.0000     0.0000
clock source latency                                 0.0000     0.0000
sdr_clk (in)                                         0.0000     0.0000 r
sdr_clk_iopad/PAD (pc3d01)                           0.0418     0.0418 r
sdr_clk_iopad/CIN (pc3d01)                           0.6857 &   0.7275 r
I_CLOCK_GEN/sdram_clk (CLOCK_GEN)                    0.0000 &   0.7275 r

. . .
```

Question 8. What symbol is now used beside the boundary net delays in **report_timing**?

The delays are now zero and have a "*" symbol indicating they are annotated delays. The hold violations are larger.

```
pt_shell> set_annotated_delay 0 -net -to [all_outputs]
pt_shell> set_annotated_delay 0 -net -from [all_inputs]
pt_shell> !report
. . .
Point                                                  Incr       Path
-----------------------------------------------------------------------
clock SDRAM_CLK (rise edge)                          0.0000     0.0000
clock source latency                                 0.0000     0.0000
sdr_clk (in)                                         0.0000     0.0000 r
sdr_clk_iopad/PAD (pc3d01)                           0.0000     0.0000 r
sdr_clk_iopad/CIN (pc3d01)                           0.6857 &   0.6857 r
I_CLOCK_GEN/sdram_clk (CLOCK_GEN)                    0.0000 &   0.6857 r

. . .

sdram_DQ_iopad_0/PAD (pc3b05)                        1.6088     3.0887 r
sd_DQ[0] (inout)                                     0.0000 *   3.0887 r
data arrival time                                               3.0887

clock SD_DDR_CLK (rise edge)                         0.0000     0.0000
clock network delay (propagated)                     3.4646     3.4646
clock reconvergence pessimism                        0.0000     3.4646
output external delay                                0.1000     3.5646
data required time                                              3.5646
-----------------------------------------------------------------------
data required time                                              3.5646
data arrival time                                              -3.0887
-----------------------------------------------------------------------
slack (VIOLATED)                                                0.4758
```


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Summary Reports for All Violations

`report_analysis_coverage`	Total violations
`-status violated`	Details on violations
`-check {setup hold}`	For these timing checks
`-sort_by slack`	Default
`-sort_by check_type`	
`report_constraint -all`	Sort violations by clock group
`report_global_timing`	
`report_qor -only_violated`	

`report_timing` Switches: Control Which and How Many

`-nworst`	Number of paths to consider per endpoint.
`-max_paths`	Number of paths to generate per design.
`-delay`	Setup (`max`) or hold (`min`) with a specific end point data transition (`max_rise`, `max_fall`, `min_rise`, `min_fall`).
`-group`	Focus on this path group (e.g. capture clock name).
`-group [get_path_group *]`	
`-to -rise_to -fall_to`	
`-from -rise_from -fall_from`	
`-through -rise_through -fall_through`	
`-exclude -rise_exclude -fall_exclude`	
`-slack_lesser_than`	
`-slack_greater_than`	

`report_timing` Switches: Control Details

`-input_pins`	Show input pins and separate net and cell delays.
`-nets`	Show net names and fanout.
`-derate`	Show derate factor for each delay.
`-path full_clock`	Show cells on clock network for propagated clocks.
`-path short`	Remove cells on data path.
`-path end`	Report endpoint, data arrival, required and slack only.

Helpful PrimeTime Commands

```
restore_session <unix_dir>
# Save the session before restoring
save_session <unix_dir>
report_timing -help
help -verbose report_timing
man report_timing
help report*; # Find commands
printvar *sig*; # Find variables
apropos clock ; # Search command DB
page_on; # An alias
page_off; # An alias
history
list_libraries; list_designs
report_lib <lib_name>; # Time units
remove_design -all
remove_lib -all
sh_list_key_bindings
set_app_var sh_line_editing_mode emacs
```

PrimeTime JOB AID

Timing Reports

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Tcl Examples

```
# This is a comment
# No need to type entire command or switch names
# Use backslash to continue lines
# Use square brackets to embed commands
# Use get_clocks to refer to clock objects
report_timing -fall_from [get_clocks PCI_CLK] \
    -delay min -max 4
# Use set_app_var to modify variables
set_app_var sh_enable_page_mode true
```

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Run Scripts

```
# .synopsys_pt.setup
alias page_on {set_app_var sh_enable_page_mode true}
alias page_off {set_app_var sh_enable_page_mode false}
alias un_on {set_app_var timing_report_unconstrained_paths true}
alias un_off {set_app_var timing_report_unconstrained_paths false}
suppress_message {ENV-003 PTSR-004}
suppress_message CMD-029
history keep 200
set_app_var sh_enable_line_editing true
foreach _file [glob -nocomplain ./tcl_procs/*.tcl] {source $_file}
```

Tcl Procedures	SolvNet Doc ID
aa; # always ask for both commands and variables	012959
view; # Pop-up window for long reports	014947

```
# variables.tcl
set_app_var report_default_significant_digits 4
set_app_var link_create_black_boxes false
set_app_var sh_source_uses_search_path true
set_app_var sh_script_stop_severity E
set_app_var timing_update_status_level high
```

Find Non-Default Application Variables

```
report_app_var -only_changed_vars
```

Path-Based Analysis

```
report_timing -pba_mode path …
report_timing -pba_mode exhaustive …
```

```
# RUN.tcl
start_profile
set_app_var search_path {. ./scripts ./libs ./designs}
lappend link_path mytech_lib.db RAM_lib.db
source -echo -verbose ./scripts/variables.tcl
read_verilog {top.v A.v B.v}
link_design TOP
read_parasitics ORCA.SPEF.gz
report_annotated_parasitics
source -echo -verbose constraints.tcl
redirect -tee ./EW.log {check_timing}
report_analysis_coverage
save_session top_savesession
redirect -tee -append ./EW.log {print_message_info}
redirect -tee -append ./EW.log {quit}
```

```
unix% pt_shell -f RUN.tcl | tee -i run.log
# If the script terminates, fix errors
# If script does not terminate, search run.log for Warnings
# Generate man pages on message ID for further information
```

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Clocks and More

```
# Get to know the design clocks
report_clock
sizeof_collection [all_clocks]
sizeof_collection [get_generated_clocks *]
rpt_clock_ports; Tcl procedure in lab, Identify ports with defined clocks
report_clock -skew SD_DDR_CLK; # Reports generated clock source latency
report_timing -path full_clock_expanded; # Expands generated clock source latency
check_timing -verbose -override clock_crossing
set_app_var timing_all_clocks_propagated true; # Clocks created as propagated
# Identify clock network start points
reset_design
clk_strt -verbose; # Tcl procedure in lab
# Flag multiple clocks propagating (suggests a missing clock controlling signal)
lappend timing_check_defaults ideal_clocks; # For check_timing
report_case_analysis; # Report all case values set in design
```

Tcl Procedure	SolvNet Doc ID
rpt_case_value	009400
get_clock_ports	012739

```
# Back Annotation and Analysis Types
report_design; # Validate the analysis type
read_parasitics -syntax_only
report_annotated_parasitics
set_annotated_delay;
remove_annotated_delay; remove_annotated_check
set_load; set_resistance; # set net RC
report_delay_calculation
set_app_var timing_remove_clock_reconvergence_pessimism true
```

Symbol	Annotation
H	Hybrid annotation
*	SDF back-annotation
&	RC network back-annotation
$	RC pi back-annotation
+	Lumped RC
<none>	Wire-load model or none

```
# Other
report_timing -group **async_default**; # Recovery/Removal
report_timing -group **clock_gating_default**
report_clock_timing -type skew
report_exceptions <-ignored>
report_min_pulse_width
report_timing -trace_latch_borrow
```

Navigation

```
all_inputs -clock CLK1 ;# Input ports clocked by CLK1
get_pins -of_object U1 ;# Pin names of U1
all_connected U1/Z ;# Nets connected to pin
all_registers -level_sensitive ; # Latches
```


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Debugging Constraints

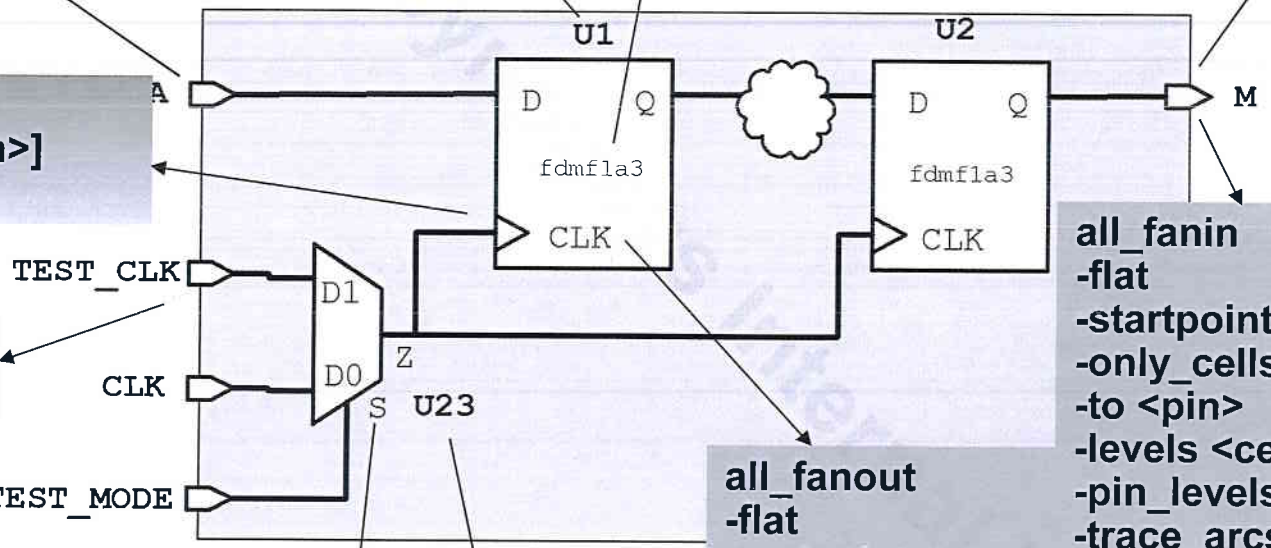
check_timing
report_analysis_coverage

report_port
-input_delay
<port>

report_cell
-connections
-verbose
<cell>

report_lib
-timing_arcs
<library>
<lib cell>

report_port
-output_delay
<port>



get_attribute
[get_pins <pin>]
clocks

report_clock
<clock>

report_case_propagation
<pins or ports>

Tcl procedure available
on SolvNet

report_disable_timing
<cells>

all_fanout
-flat
-endpoints
-only_cells
-from <pin>
-levels <cell#>
-pin_levels <pin#>
-trace_arcs enabled

all_fanin
-flat
-startpoints
-only_cells
-to <pin>
-levels <cell#>
-pin_levels <pin#>
-trace_arcs enabled

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